

The Macro Alibi: Subjective Risk Attribution in Analyst Scenarios

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May 13, 2026

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Abstract

We study how analysts attribute risk in scenario-based valuation reports. Using sell-side analysts' valuations and narratives, we document Macro Bear Bias: analysts over-attribute downside risk to macroeconomic forces. Bear-case narratives emphasize macro risk far more than base- or bull-case narratives, even though realized CAPM R^2 is similar across realized bear, base, and bull outcomes conditional on the same market state. This bias matters economically: the bear–base macro-attention gap positively predicts subsequent base-case forecast errors, and portfolios formed on a nonlinear bias-adjusted signal earn monthly CAPM alphas of up to 1.89%. Mechanism tests are consistent with an availability-heuristic channel—analysts rely on salient macro-crisis templates when describing downside risk—while career-concern incentives proxies carry little explanatory power.

Keywords: macro bear bias, narrative finance, downside risk, analyst beliefs, large language model

JEL: C55, G11, G12, G14, G23

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1 Introduction

Understanding how market participants perceive risk is a central question in finance. Asset prices depend not only on fundamentals, but also on how investors and financial intermediaries interpret the source and meaning of uncertainty. Yet there is still limited evidence on subjective perceptions of risk in real-world market settings. A major reason is data limitation. Much of the existing evidence comes from surveys, which are informative but often coarse, infrequent, and detached from consequential market communication. Such data make it difficult to observe how agents explain risk, how they decompose uncertainty into different sources, and how those perceptions vary across scenarios.

Our paper brings to bear much richer information for studying risk perceptions by combining better data and better methods. On the data side, we use Morgan Stanley’s Risk-Reward Framework reports from 2007 to 2025, which require analysts to articulate bull, base, and bear scenarios together with narrative justifications. It makes analysts’ subjective views unusually transparent. Rather than observing only a point forecast, we observe how analysts describe upside and downside states, which risks they emphasize in each scenario, and how they verbally attribute the sources of potential outcomes. The structure of these reports therefore allows us to move beyond stated forecasts and examine the narrative architecture underlying perceived risk across scenarios. On the method side, we use large language models to systematically extract and classify the risks embedded in those narratives.

Together, these settings allow us to study a basic but underexplored question: how do analysts subjectively perceive and narrate risk across scenarios, and do those narratives accurately reflect the underlying sources of risk? More specifically, when analysts describe downside states, do they correctly condition on the market environment, or do they systematically rely on a macro-dominant narrative template that overstates the role of aggregate macroeconomic risk in bad outcomes? This question is important for at least three reasons. First, whether downside risk is perceived as macro or firm-specific directly affects valuation, discount rates, and expected returns. Second, analysts are influential information intermediaries, so distortions in the way they frame risk can propagate

into investor beliefs and market prices. Third, understanding whether downside narratives are well-calibrated helps distinguish rational state-contingent forecasting from narrative distortion and therefore sheds light on how beliefs are formed, communicated, and translated into forecast errors and asset-pricing outcomes.

Our central finding is that analysts perceive downside risk through a systematically distorted narrative lens: bear-case narratives place substantially greater emphasis on macroeconomic risks than base- or bull-case narratives. This asymmetry is not merely a difference in pessimistic tone. It reflects a distinct pattern of subjective risk attribution. To assess whether this pattern is justified, we compare ex ante narrative attribution with an ex post benchmark based on realized CAPM R^2 over the year following report issuance, conditional on the same market environment. If downside outcomes are truly more macro-driven, realized return variation should be more strongly explained by market movements in realized bear cases than in realized base or bull cases. Instead, we find that conditional on the same subsequent market state, realized CAPM R^2 is similar across realized bear, base, and bull outcomes, even though bear narratives remain persistently macro-heavy. We refer to this distortion as **Macro Bear Bias**: the tendency to over-attribute downside outcomes to aggregate macroeconomic risks in bear-case narratives.

This finding contributes to the literature in two ways. First, we show that understanding subjective risk perception requires studying narratives, not just forecasts. Numerical targets reveal what analysts expect, but narratives reveal how they map uncertainty into explanations. That mapping is economically meaningful because it determines whether downside risk is framed as macro, industry-specific, or firm-specific. Second, we show that the three-scenario structure of analyst reports provides a powerful way to identify miscalibration in subjective risk perception. By comparing bear, base, and bull narratives within the same report, we isolate the incremental macro attribution that analysts add when moving from the expected outcome to the downside. This form of narrative misattribution would be difficult to detect in single-scenario settings or with traditional survey data.

Scenario narratives help organize analysts' broader valuation judgments, so an excessively macro-heavy bear narrative can spill over into the base-case forecast itself. Consistent with this

mechanism, we show that the bear–base macro-attention gap predicts subsequent errors in analysts’ base-case forecasts. Firms whose downside narratives place unusually heavy emphasis on macro risk subsequently exhibit more positive forecast errors, indicating that analysts were too pessimistic *ex ante*. Moreover, this predictive relation is strongly nonlinear. Flexible nonlinear methods extract substantially more forecasting power than linear models, suggesting that the effect of Macro Bear Bias depends on how downside narratives interact with the prevailing market environment. In this way, the paper links subjective risk perception, narrative risk attribution, and economically meaningful forecast errors.

The forecast-error evidence also translates into portfolio performance. We use the nonlinear model to construct a bias-adjusted forecast that adds the model-predicted forecast-error correction to the analyst’s raw base-case expected return. This adjustment has substantial return relevance. Sorting on analysts’ raw base-case forecasts produces no meaningful abnormal returns, suggesting that the base target alone does not provide an exploitable cross-sectional signal. By contrast, sorting on the nonlinearity-adjusted forecast generates large and statistically significant long–short returns. The value-weighted strategy earns an average monthly return of 2.01%, a CAPM alpha of 1.89%, and an annualized Sharpe ratio of 1.09; the equal-weighted strategy earns an average monthly return of 1.79%, a CAPM alpha of 1.73%, and an annualized Sharpe ratio of 0.99. These results show that Macro Bear Bias is not only a narrative distortion or a forecast-error pattern; it also contains economically important information for return prediction.

We then study the mechanisms behind Macro Bear Bias. The evidence points primarily to a cognitive mechanism: availability heuristic. When analysts construct bear scenarios, they appear to draw disproportionately on salient macro-crisis episodes—especially the Global Financial Crisis and the COVID-19 crash. These episodes make macro-driven collapses especially easy to recall, causing analysts to use a macro-crisis template when describing downside risk. Under this interpretation, bear narratives are not a rational response to contemporaneous macro conditions. Rather, analysts over-apply a readily available crisis narrative to downside states, even when the realized source of risk is not especially macro in nature.

Several pieces of evidence support this availability heuristic mechanism. First, bear-case macro

attention responds strongly to negative macro news shocks. Negative macro topics enter bear-case narratives quickly when they become salient in the news, but decay slowly after media attention subsides. More specifically, unexpected increases in WSJ attention to macro topics predict greater analyst attention to those same topics in bear-case narratives over the following months, especially when the associated news sentiment is negative. This pattern suggests that salient negative macro news makes macro explanations more cognitively available when analysts construct downside scenarios.

Second, the effect is stronger in sectors where crisis-based macro narratives are especially salient. After the Global Financial Crisis, high-systematic-risk industries such as financials, energy, and materials exhibit substantially larger bear-base macro-attention gaps, consistent with analysts applying a financial-contagion template to sectors most visibly associated with the crisis.

Third, analyst-level crisis exposure matters. Analysts who experienced the Global Financial Crisis while working at Morgan Stanley exhibit larger bear-base macro-attention gaps, whereas broader profession-wide crisis experience has little explanatory power. This pattern suggests that availability operates not only through personal memory of the crisis, but also through institution-specific narrative conventions formed during salient macro episodes.

Fourth, the bear-base macro-attention gap co-moves within sector-quarter peer groups. An analyst's own bear-base macro-attention gap is positively related to the leave-one-out average gap among other analysts covering firms in the same sector and quarter. This sector-level narrative convergence is consistent with analysts drawing on shared pools of salient macro-crisis templates through common media exposure, peer discussion, and industry-specific information environments.

Collectively, these four findings indicate that analysts appear to anchor bear-case narratives to macro-crisis templates that are vivid, recently activated, institutionally reinforced, and shared within sector-level information networks. Once these templates become salient, they remain available as default explanations for downside risk, leading analysts to over-attribute bear-case outcomes to macro forces rather than fully recalibrating the source of risk in each firm-specific setting.

The availability heuristic also helps explain why Macro Bear Bias affects base-case forecasts. Analysts' bull, base, and bear cases are not independent forecasts; they are components of a coherent

valuation narrative. Once a salient macro-crisis template is used to describe the bear case, that framing can spill over into the base case by making aggregate risk appear more important for the firm's valuation. In a discount-rate interpretation, the macro-heavy bear narrative raises the analyst's perceived exposure to systematic risk, increasing the implicit discount rate applied to base-case cash flows and depressing the base-case target. Thus, the availability heuristic generates the initial over-attribution of downside risk to macro forces, while narrative coherence and discount-rate contamination transmit this bias into economically meaningful forecast errors.

We also evaluate a strategic career-concerns explanation. Under this view, macro-based downside narratives may be reputationally safer than sharp firm-specific negative claims because macro explanations are harder to falsify, less damaging to management relationships, and more likely to be shared by other market participants. However, the data provide little support for this explanation. Junior analysts do not display stronger Macro Bear Bias; firms with recent Morgan Stanley investment-banking relationships do not exhibit significantly larger macro gaps; and analyst-level proxies for career incentives explain little of the variation in the bias. These results suggest that Macro Bear Bias is better understood as an availability-driven cognitive distortion, amplified through narrative organization, rather than as a deliberate strategic communication device.

More broadly, the paper studies whether one of the market's key information producers correctly understands and communicates the driver of downside risk. That question matters for belief formation, price discovery, and the role of narratives in financial markets. By combining a uniquely informative setting with LLM-based measurement, we provide new evidence on how sell-side analysts perceive risk, how those perceptions become embedded in financial narratives, and how narrative distortions translate into forecast errors and return predictability.

Literature

Our paper connects to four strands of literature. [Ang and Chen \(2002\)](#) document that correlations between U.S. equities and the market are substantially higher for downside moves than for upside moves, and [Ang, Chen, and Xing \(2006\)](#) establish a downside beta premium in the cross-section of stock returns. [Lettau, Maggiori, and Weber \(2014\)](#) show that a conditional CAPM in which

betas are elevated in bad states prices the cross-section of currency and equity returns, extending the partial rationality of state-dependent risk pricing to multiple asset classes. [Lewellen and Nagel \(2006\)](#) caution, however, that the conditional CAPM does not mechanically resolve standard asset-pricing anomalies, since beta variation would need to be implausibly large. [Lewellen, Nagel, and Shanken \(2010\)](#) further show that the standard cross-sectional testing approach is poorly suited to discriminating among conditional factor models. [Kacperczyk, Van Nieuwerburgh, and Veldkamp \(2016\)](#) and [Kacperczyk, Nieuwerburgh, and Veldkamp \(2014\)](#) show that fund managers optimally shift attention between aggregate market-timing and firm-specific stock-picking as a function of the business cycle: macro attention is concentrated in recessions, when systematic risk is most relevant, and dispersed in expansions. Our paper contrasts with this optimal attention-allocation result: analysts do not make the analogous shift in bear-case narrative construction. Bear narratives remain macro-heavy across all market regimes, suggesting that the narrative template is set at the time of scenario drafting without subsequent conditioning on realized states. These regularities constitute the genuine empirical foundation that analysts overgeneralize when constructing bear-case narratives. We complement this literature by characterizing the limits of that generalization: the empirical regularity in downside comovement is state-dependent, and analysts apply macro-heavy bear narratives unconditionally even when the realized evidence calls for state-dependent conditioning. The macro bear bias is not a departure from the Ang-Chen fact—it is the failure to condition on it.

A second strand studies the economic narratives analysts construct to justify their forecasts—the qualitative reasoning that accompanies and shapes quantitative outputs.¹ [Joos, Piotroski, and Srinivasan \(2016\)](#) show that analyst-constructed risk scenarios contain information about fundamental uncertainty not captured by standard quantitative forecasts. [Ke \(2024\)](#) use analyst

¹A complementary and burgeoning literature studies subjective beliefs embedded in analyst *numerical* forecasts of earnings and long-term growth. [La Porta \(1996\)](#) shows that stocks with high analyst long-term growth forecasts earn negative abnormal returns, an early indication that analyst growth expectations are systematically optimistic. [Bordalo et al. \(2019\)](#) and [Bordalo et al. \(2024\)](#) develop the diagnostic expectations model and show it accounts for the overreaction pattern in analyst earnings forecasts and the associated stock return anomalies. [Bouchaud et al. \(2019\)](#) document that analysts are slow to revise earnings expectations in response to profitability news, generating the profitability anomaly. [De la O and Myers \(2021\)](#) use analyst forecasts to measure subjective cash flow and discount rate expectations, finding that cash flow news drives return variation more than standard rational-expectations models imply.

narratives to recover subjective return beliefs, documenting that stated reasoning carries information beyond point estimates. [Bradshaw et al. \(2025\)](#) analyze qualitative signals in analyst reports at scale, showing that textual content unlocks dimensions of information not reflected in price targets. [Bastianello, Décaire, and Guenzel \(2024\)](#) show that forecasters with different internal models of the economy form persistently divergent expectations. We extend this literature by exploiting the *across-scenario* variation in narrative content—bear versus base versus bull—rather than the level of any single scenario. The three-scenario structure of the Morgan Stanley reports allows us to identify the incremental macro attribution that analysts add when moving from the expected outcome to the downside, which is precisely the miscalibration of interest and which a single-scenario analysis cannot detect.

Our paper also contributes to the emerging literature on narrative finance. [Shiller \(2019\)](#) argues that viral economic narratives—most potently, recession narratives—shape agents’ expectations and drive real outcomes. Our evidence complements this view: “recession risk” functions as a default bear-case template in analyst reports, applied regardless of whether the underlying downside risk is macro-driven. [Addoum, Meursault, and Murfin \(2024\)](#) document macro emphasis in debt investor narratives for corporate borrowers, providing complementary evidence from the bond side of the capital structure. We contrast with their level-based analysis by establishing that the equity bear narrative constitutes a state-dependent miscalibration: macro emphasis does not co-vary with the fraction of bear-case realizations that are genuinely systematic, which a level analysis cannot uncover.

Finally, our findings connect to the career concerns and herding literature in finance and economics. [Holmström \(1999\)](#) establishes the canonical model in which agents distort actions to signal ability, and [Scharfstein and Stein \(1990\)](#) show that career incentives lead agents to mimic each other even when private information would call for deviation. In the analyst context, [Hong and Kubik \(2003\)](#) show that analysts early in their careers issue more optimistic forecasts to manage the reputational cost of bold pessimistic calls, and [Chevalier and Ellison \(1999\)](#) document the same incentive structure for mutual fund managers. We reconcile this literature with the macro bear bias: attributing bear-case downside to macroeconomic forces shifts the perceived locus of the call

from the analyst’s own judgment to an external macro shock, reducing reputational exposure if the forecast fails. This incentive to externalize blame complements the availability heuristic as a mechanism for why the macro bear bias is stable across firms, market regimes, and analyst cohorts.

The rest of the paper proceeds as follows. Section 2 documents Macro Bear Bias: we describe the data, construct the narrative attention measures, and show that bear-case macro attention is persistently elevated relative to the realized CAPM R^2 benchmark. Section 3 develops the conceptual framework, formalizes the bias, and presents empirical evidence—both linear and nonlinear—on forecast error prediction. Section 4 tests mechanisms empirically: the availability heuristic receives cross-sectional support from industry–crisis interactions and sector-level narrative convergence, and a narrative coherence channel is identified through directional boldness; the career-concerns proxies we observe show little explanatory power. Section 5 concludes.

2 Documenting Macro Bear Bias

This section documents the paper’s main empirical result: Macro Bear Bias, the tendency for analysts to over-attribute downside outcomes to macroeconomic drivers in bear-case narratives. Analysts do not describe upside and downside risks symmetrically. Rather, bear scenarios receive disproportionately greater macro emphasis than base or bull scenarios. We begin by introducing the Morgan Stanley Risk-Reward reports, which provide an unusually rich setting for studying subjective risk attribution. We then use the narrative text in these reports to measure how analysts allocate attention across macro, industry, and firm-level drivers in bull, base, and bear cases. Finally, we compare these ex ante narrative attributions with an ex post realized benchmark based on CAPM R^2 , conditional on the same market environment. This comparison allows us to assess whether the macro-heavy framing of bear scenarios reflects the true driver of downside outcomes or instead a systematic misattribution. The evidence points clearly to the latter: the persistent excess macro emphasis in bear narratives is the empirical signature of Macro Bear Bias.

2.1 Data

Our paper uses data from the Risk-Reward Framework in Morgan Stanley analyst reports. In total, we collected 63,191 Risk-Reward reports from January 2007 to May 2025.

Traditional sell-side research often relied on single-point price targets or coarse recommendations (e.g., “buy,” “hold,” or “sell”), which can convey a false sense of precision and provide limited information about the range of possible future outcomes. To address this limitation, Morgan Stanley introduced the Risk-Reward Framework in 2007, requiring analysts to present and justify 12-month stock-price expectations under three scenarios: bull, base, and bear. This framework was designed to provide a more informative and differentiated research product for institutional investors, who care not only about expected return but also about the distribution of potential outcomes and the sources of upside and downside risk. By explicitly presenting upside and downside scenarios, these reports help investors assess the skewness of expected payoffs, understand the assumptions underlying each case, and incorporate analyst views more directly into portfolio decisions. The reports are particularly valuable for our purposes because they provide both structured scenario-specific price targets and narrative discussion of the reasoning, key drivers, and investment debate behind those targets. For introductions to the framework, see [Weyns et al. \(2011\)](#) and [Srinivasan and Lane \(2011\)](#).

This dataset offers two main advantages. First, in addition to the base-case forecast, it contains analyst expectations under both bull and bear scenarios, which allows us to measure subjective upside and downside risk expectations separately. Second, the accompanying narratives explain the assumptions and reasoning behind each scenario. Using natural language processing (NLP) and large language models (LLMs), we can systematically extract and analyze these texts to identify the drivers of perceived upside and downside risks.

Table 1 reports the number of available Risk-Reward reports and the number of covered firms by year. Because the Risk-Reward Framework is specific to Morgan Stanley and covers primarily large-capitalization, analyst-covered U.S. equities, the results should be interpreted with this sample scope in mind.

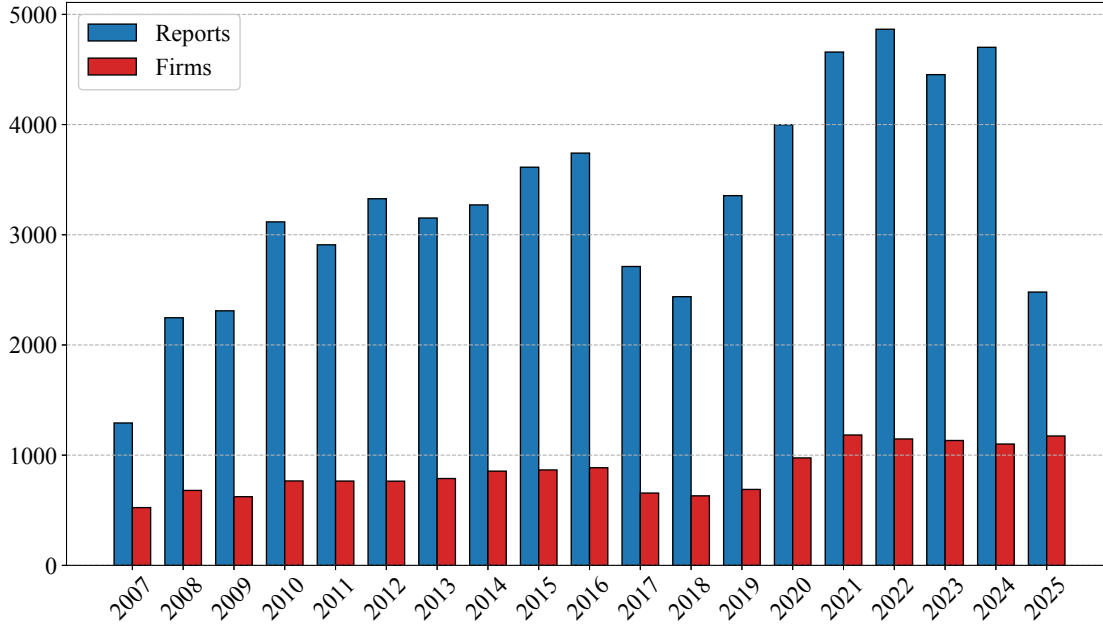


Figure 1 Histogram of Reports and Firms by Year.

Note: This figure shows the yearly count of Morgan Stanley Risk-Reward analyst reports and covered firms from January 2007 to May 2025.

2.2 Textual Identification of Subjective Risks

We focus on the narrative text in each Morgan Stanley Risk-Reward report to understand how analysts justify their bull-, base-, and bear-case expectations. Specifically, we examine whether analysts exhibit a downside–upside asymmetry in the types of risks they emphasize across scenarios, distinguishing among macro-level, industry-level, and firm-level risks. To quantify the content of these narratives, we apply the GPT prompt described in Appendix Section A using the gpt-4o-mini model. Our main finding is that analysts place significantly greater emphasis on macro-level risk in bear scenarios than in base or bull scenarios. As a robustness check, we also examine the distinction between systematic and idiosyncratic risk using the prompt described in Appendix Section A.3.²

Figure 2 presents the report-level distributions of risk attention across scenarios. Panel A

²Text-share attention is our proxy for subjective risk attribution, not a literal measurement of it. A passage that mentions a macro factor in order to dismiss it is counted as macro attention by the classifier even though the analyst’s substantive attribution runs to firm-specific resilience. To the extent dismissive mentions are present in bear-case narratives, the measured bear–base macro gap understates the true attribution gap when the dismissals are concentrated in bear cases, and overstates it when concentrated in base or bull cases. We view the proxy as conservative for the paper’s claim because bear-case passages are most likely to invoke macro factors substantively rather than dismissively.

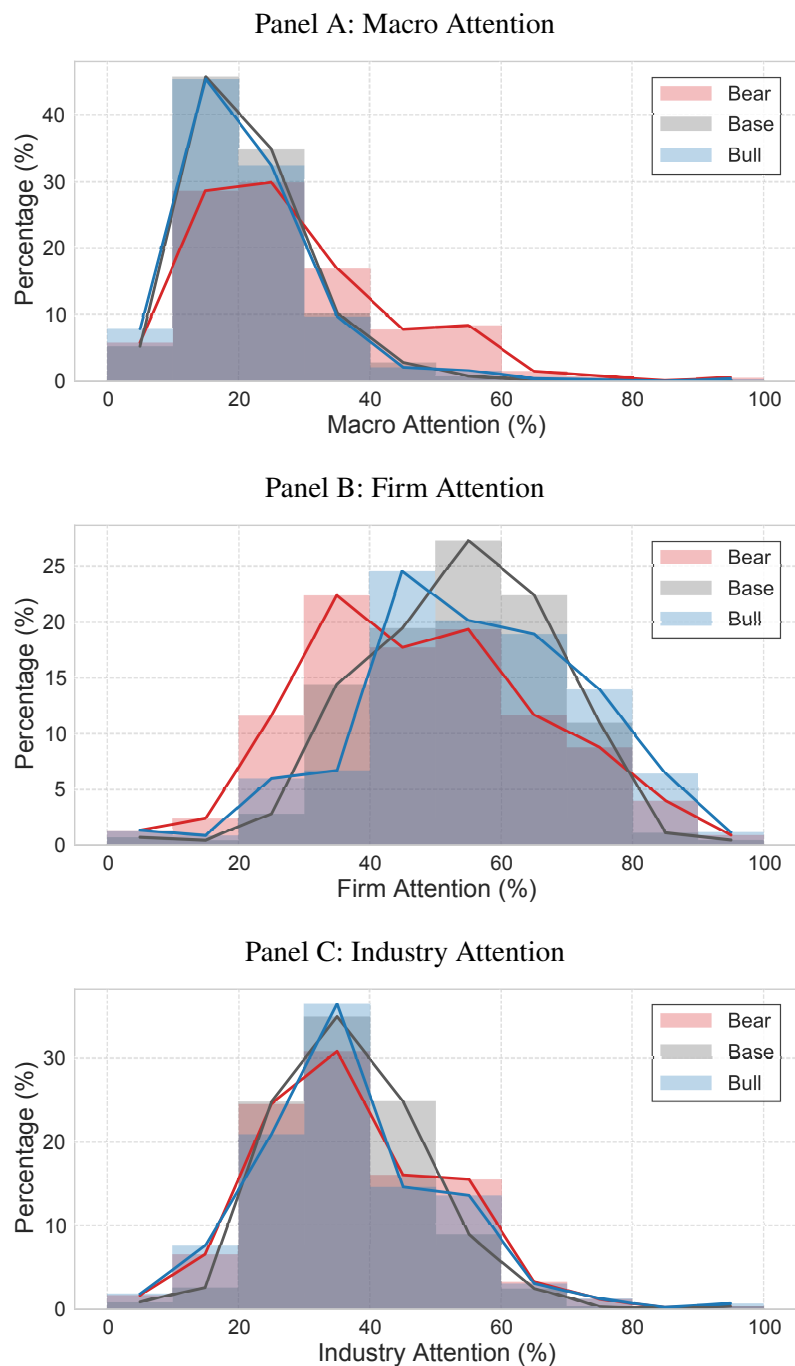


Figure 2 Attention Distribution to Macro, Firm, and Industry News.

Note: Report-level distribution of attention shares to firm-, industry-, and macro-level news across bull, base, and bear cases.

shows the distribution of attention to macro-level risks in bull, base, and bear cases. Panel B reports the corresponding distribution for firm-level risks, and Panel C does the same for industry-

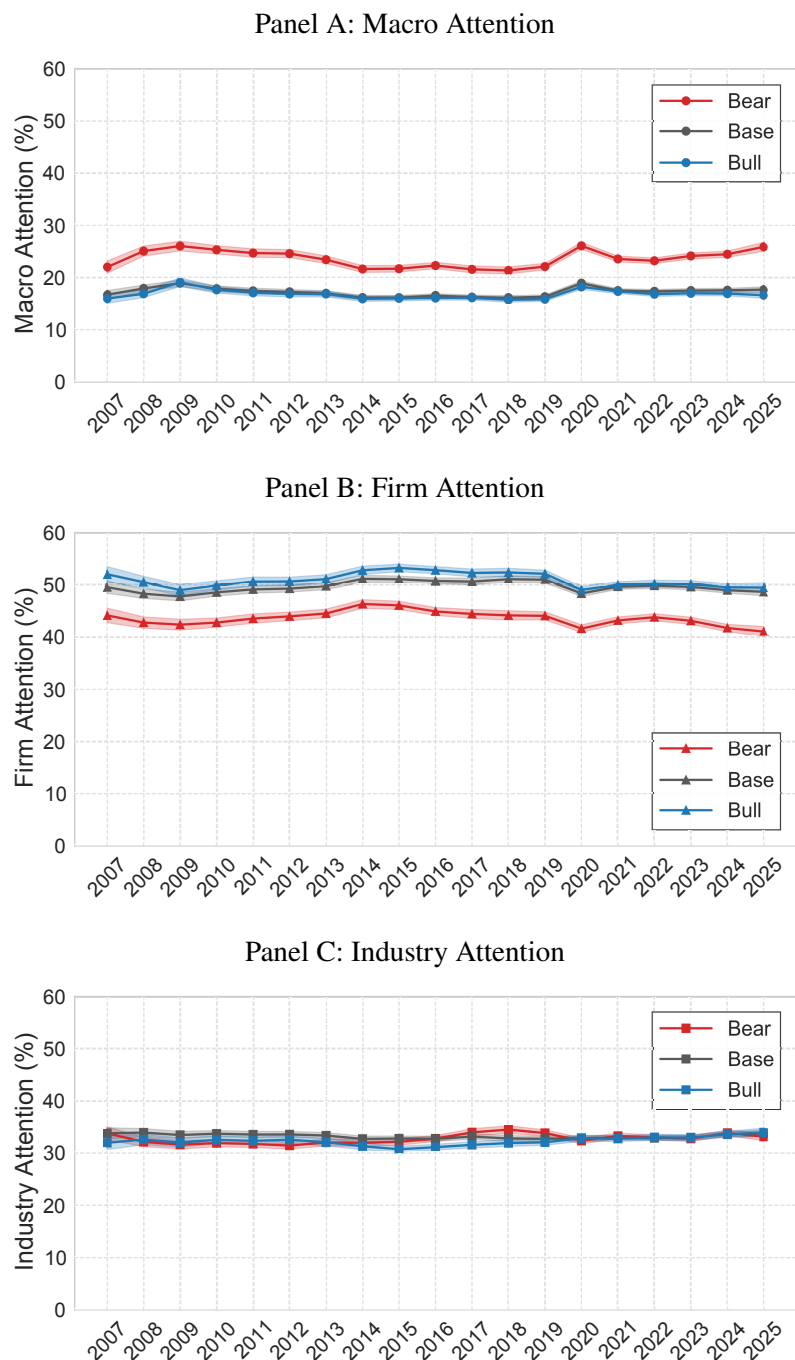


Figure 3 Time Series of Average Attention to Macro, Firm, and Industry News.

Note: Time series of report-level annual average attention shares to macro-, firm-, industry-level news. The shaded areas are 99% confidence interval of the average.

level risks. The figure reveals a clear pattern: relative to bull and base scenarios, bear scenarios assign substantially greater weight to macro-level risks and significantly less weight to firm-level

Table 1 Scenario Dummies and Attention Shares

Note: This table reports panel regressions of attention shares on scenario indicators. The dependent variables are macro attention (cols 1–2), firm attention (cols 3–4), and industry attention (cols 5–6). Base scenario is the omitted category; the constant reports the mean attention share for the Base scenario. Bear and Bull coefficients measure the difference relative to Base. All specifications include firm fixed effects; even columns add year-month fixed effects. Standard errors are clustered by firm and month. t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Macro	Macro	Firm	Firm	Industry	Industry
Bear	6.4687*** (34.09)	6.4687*** (34.07)	-6.0672*** (-31.54)	-6.0672*** (-31.53)	-0.4027** (-2.55)	-0.4027** (-2.55)
Bull	-0.3982*** (-5.54)	-0.3982*** (-5.54)	1.1609*** (10.20)	1.1609*** (10.19)	-0.7654*** (-7.21)	-0.7654*** (-7.21)
Constant	17.1843*** (443.27)	17.1843*** (443.27)	49.6454*** (858.57)	49.6454*** (858.57)	33.1654*** (722.96)	33.1654*** (722.96)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE		Yes		Yes		Yes
Observations	187,917	187,917	187,917	187,917	187,917	187,917
Within R^2	0.0781	0.0863	0.0431	0.0521	0.0006	0.0053

risks. By contrast, the distributions of macro-level attention in bull and base scenarios are similar to each other, as are the distributions of firm-level attention across these two scenarios. Industry-level attention shows broadly similar distributions across all three scenarios.

Figure 3 complements this distribution evidence by showing the time-series evolution of average attention to macro-, firm-, and industry-level risks from 2007 to 2025. Panel A plots macro-level attention, Panel B plots firm-level attention, and Panel C plots industry-level attention. The time-series evidence reinforces the distributional results: throughout the sample period, analysts consistently devote more attention to macro-level risks in bear scenarios than in bull or base scenarios, while placing less attention on firm-level risks in bear scenarios. In contrast, attention patterns in bull and base scenarios remain broadly similar over time for both macro- and firm-level risks. Industry-level attention again appears relatively stable and similar across bull, base, and bear cases. For completeness, Figure A.1 reports the distribution at each point in time.

2.3 Realized CAPM R^2

The evidence so far establishes that analysts assign substantially more macro attention to bear scenarios than to base or bull scenarios. But is this differential attention warranted? To assess whether analysts are correct to place greater weight on macro risks in bear scenarios, we construct an *ex post* realized benchmark based on the realized CAPM R^2 in each case. If macro risks truly matter more in downside states, a firm's realized return variation should be more strongly explained by market-wide movements when the report ultimately realizes a bear outcome than when it realizes a base or bull outcome. We test whether the realized CAPM R^2 is significantly higher for realized-bear reports than for realized-base or realized-bull reports. A higher realized R^2 in bear outcomes would support analysts' macro emphasis in downside scenarios; otherwise, it would suggest overweighting.

Our empirical procedure has three steps. First, for each report, we compute the realized *ex post* CAPM R^2 by regressing the firm's daily excess returns on daily market excess returns over the 252 trading days following the report release. This R^2 measures the extent to which the firm's subsequent stock return variation is explained by systematic market movements.

Second, we classify each report as realized-bear, realized-base, or realized-bull according to the relationship between the analyst's target prices and the stock's realized price one year after the report release. A report is classified as realized-bear if the realized price is less than or equal to the bear-case target price. It is classified as realized-bull if the realized price is greater than or equal to the bull-case target price. If the realized price lies strictly between the bear and bull targets, the report is classified as realized-base.

Third, we aggregate these report-level realized R^2 measures to the firm level. For each firm, the realized-bear R^2 is the average realized CAPM R^2 across all of its realized-bear reports. Likewise, the realized-base and realized-bull R^2 are defined as the average realized CAPM R^2 across the firm's realized-base and realized-bull reports, respectively. Constructing the measures at the firm level allows us to compare whether the realized R^2 is higher in bear outcomes than in base or bull outcomes for the same firm, thereby reducing concerns that cross-firm differences drive the results.

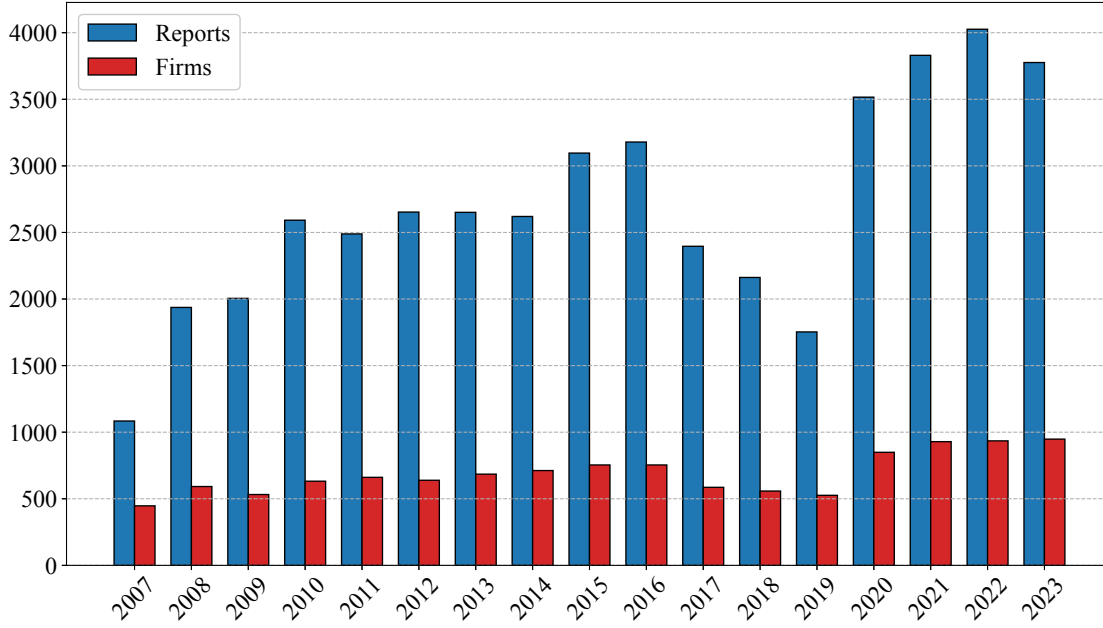


Figure 4 Yearly Distribution of Reports and Firms after Matching with Returns.

Note: This figure shows the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms after matching the reports with one-year-ahead firm returns.

To make these comparisons within a common market environment, we further construct firm-level realized R^2 measures separately by market condition. Specifically, under a given market state, we restrict the sample to firms that experience all three realized report types—bear, base, and bull—within that same market condition. For example, when analyzing periods in which the market excess return over the 252 trading days after the report release lies between -5% and 5% , we include only firms that have realized-bear, realized-base, and realized-bull reports within that market-return range. This restriction ensures that comparisons across realized cases are not mechanically driven by differences in overall market environments.

We obtain stock return data from the CRSP database. After matching Morgan Stanley reports to 252-trading-day forward returns following the report release date, the sample contains 45,348 reports in total. Figure 4 presents the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms in the matched sample.

To mitigate distortions from stock splits and reverse splits, we construct an alternative measure of the one-year-ahead realized price by compounding daily returns excluding dividends, denoted

by $r_{t,x}$, over the 252 trading days after the report release:

$$\text{constructed realized price} = \text{current price} \times \prod_{t=1}^{252} (1 + r_{t,x}).$$

If the realized price in one year differs from the constructed realized price by more than 10% in either direction, we replace the observed realized price with the constructed realized price. This adjustment helps correct cases in which stock splits or reverse splits mechanically distort the reported price level.

Table 2 Summary Statistics

Note: This table reports summary statistics for the Morgan Stanley Risk-Reward report sample from 2007 to 2023. Panel A reports report-level attention shares (percentages) by scenario. Panel B reports firm characteristics at the report date. All attention shares are extracted by GPT using the prompt in Appendix A.

Variable	N	Mean	Median	Std	Min	P25	P75	Max
Panel A: Narrative Attention								
Bull: Macro	45765	16.70	10.00	11.11	0.00	10.00	20.00	100.00
Bull: Industry	45765	32.29	30.00	14.33	0.00	20.00	40.00	100.00
Bull: Firm	45765	51.01	50.00	17.32	0.00	40.00	60.00	100.00
Base: Macro	45765	17.02	15.00	9.46	0.00	10.00	20.00	100.00
Base: Industry	45765	33.14	30.00	11.49	0.00	25.00	40.00	100.00
Base: Firm	45765	49.84	50.00	14.43	0.00	40.00	60.00	100.00
Bear: Macro	45765	23.33	20.00	15.10	0.00	10.00	30.00	100.00
Bear: Industry	45765	32.64	30.00	14.06	0.00	20.00	40.00	100.00
Bear: Firm	45765	44.03	40.00	18.32	0.00	30.00	60.00	100.00
Panel B: Returns and Forecast Variables								
Return	45765	0.12	0.07	0.74	-3.29	-0.17	0.31	74.14
Subjective Return	45765	0.20	0.10	1.79	-4.67	-0.01	0.22	115.33
Forecast Error	45765	-0.08	-0.02	1.91	-115.26	-0.30	0.26	74.03
$\hat{\beta}_{\text{CAPM}}$	43008	1.13	1.08	0.44	-1.04	0.83	1.37	5.29
R^2_{CAPM}	43008	0.32	0.30	0.18	0.00	0.18	0.44	0.86

Table 2 reports summary statistics of core narrative and firm-level data. In Panel A highlights the narrative asymmetry across scenarios. Bear-case macro attention averages 23.4%, compared with 17.0% and 16.7% in the base and bull cases, respectively—a bear–base gap of approximately 6.3 percentage points. The mirror image appears in firm-level attention: bear-case firm attention

averages 44.0%, roughly 6 percentage points below the base-case mean of 49.9%, while industry attention is nearly identical across all three scenarios. Panel B shows that analyst base-case targets are slightly optimistic on average (mean forecast error of -0.08), and that the coverage universe skews toward high-beta firms (mean $\hat{\beta}_{\text{CAPM}} = 1.13$), consistent with Morgan Stanley's focus on large-cap stocks.

Figure 5 illustrates how realized bear, base, and bull reports are distributed across different market states. To characterize market conditions, we group reports according to the forward 252-day market excess return following the analyst report release. For each point on the horizontal axis, we consider reports whose forward market excess returns fall within a centered 10% window around that point.

The two panels present the same data from complementary perspectives. Panel A focuses on the composition of realized outcomes within each market state. For a given market-return state, it reports the shares of realized bear, base, and bull reports among all reports that fall into that state. This perspective shows how the relative likelihood of each realized outcome changes as aggregate market conditions become weaker or stronger. The figure shows a clear pattern: as future market excess returns rise, the share of realized bear reports declines, while the share of realized bull reports increases. In contrast, the share of realized base reports remains comparatively stable across market states. This pattern is intuitive. When the overall market performs poorly, more stocks are likely to end up near analysts' downside scenarios, making realized bear outcomes more common. When the market performs strongly, more stocks are likely to reach their upside scenarios, increasing the frequency of realized bull outcomes.

Panel B instead focuses on how each realized outcome is distributed across market states. For realized bear, base, and bull reports separately, it plots the fraction of that report type that falls into each market-return state. This perspective highlights the market environments in which each realized outcome is most likely to occur. Realized bear reports are spread more broadly across market states, resulting in a relatively flat profile. By contrast, realized base reports, and especially realized bull reports, become more concentrated in stronger market states. This pattern suggests

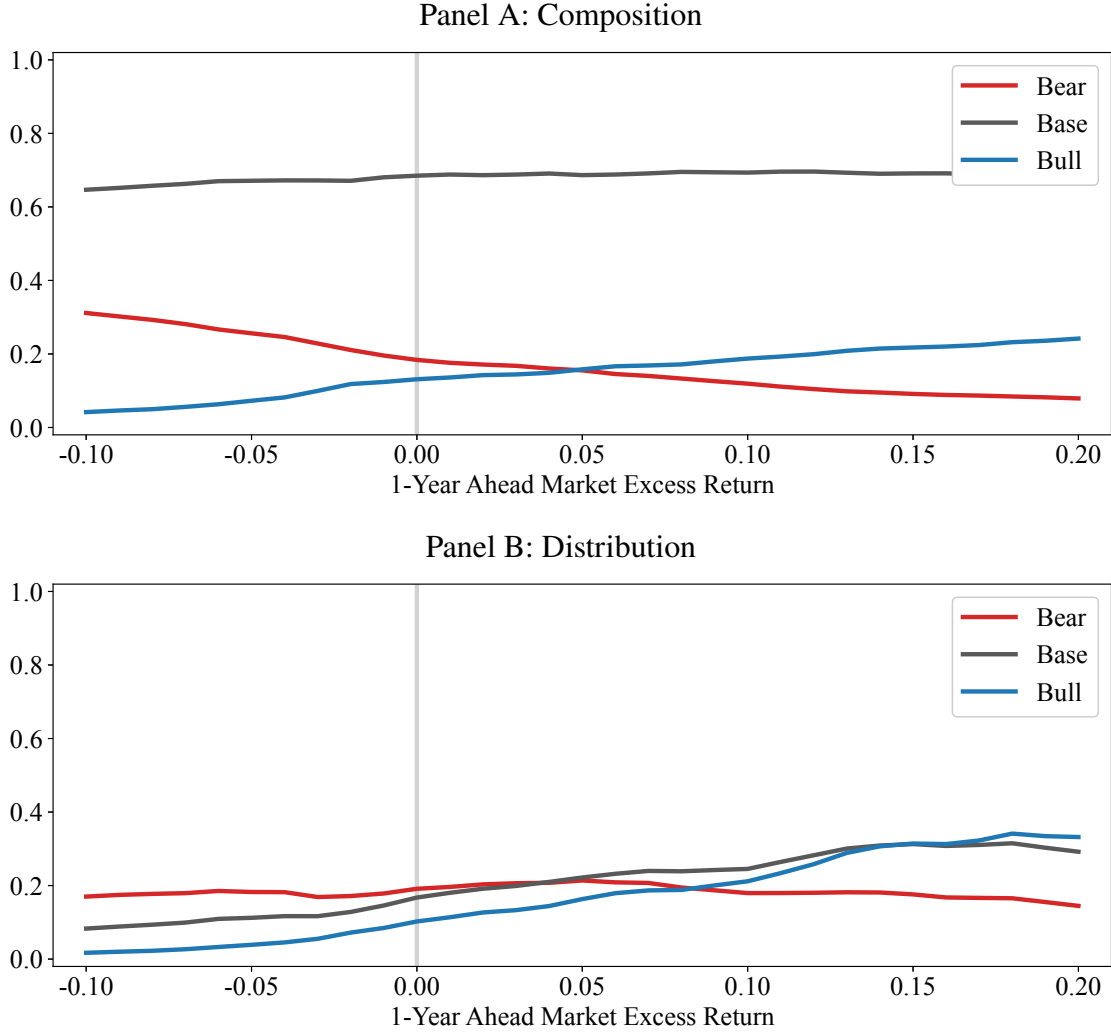


Figure 5 Composition and Distribution of Realized Bear, Base, and Bull Reports Across Market States.

Note: This figure plots the composition and distribution of realized bear, base, and bull reports across market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. Panel A shows the composition of bear, base, and bull reports within each return state:

$$\frac{\text{Number of realized bear/base/bull reports in a return state}}{\text{Total number of reports in that market excess return}}.$$

Panel B shows the distribution for realized bear, base, and bull reports:

$$\frac{\text{Number of realized bear/base/bull reports in a market excess return state}}{\text{Total number of realized bear/base/bull reports}}.$$

that the distribution of forward market excess returns is tilted toward positive values, so realized base and bull outcomes are more likely to occur when subsequent market conditions are favorable.

Read jointly, the two panels provide a more complete picture of the relation between realized report outcomes and subsequent market conditions. The figure shows that realized outcomes are systematically linked to the broader market environment: realized bear outcomes are more prevalent when subsequent market conditions are weak, while realized bull outcomes become both more common and more concentrated when subsequent market conditions are strong. This implies that, when comparing realized bear, base, and bull benchmarks, it is important to condition on market state; otherwise, observed differences may conflate firm-level realization with variation in aggregate market performance.

2.4 Empirical Evidence on Macro Bear Bias

With the realized CAPM R^2 benchmark and market-state conditioning framework in hand, we now examine whether the elevated macro emphasis in bear-case narratives reflects the actual macro content of bear-case outcomes. We show that it does not. Instead, analysts systematically overweight macro risk in bear scenarios. We refer to this distortion as Macro Bear Bias: the tendency to over-attribute downside outcomes to aggregate macro risks in bear-case narratives.

To ensure an apples-to-apples comparison with realized R^2 , we construct firm-level ex ante macro attention for each case (bear, base, and bull) as the average macro attention across all reports of the firm that fall within the same market state. Thus, the comparison with the ex post realized R^2 is based on the exact same subset of reports. Figure 6 plots realized R^2 and macro attention conditional on market state. For each point x on the horizontal axis, we consider reports whose forward 252-day market excess returns fall within a centered 10% window, $[x - 5\%, x + 5\%]$, around that point.³

Panel A shows the average CAPM R^2 over the 252 trading days after report release for realized bear (red), realized base (yellow), and realized bull (green) cases. Conditional on the same market

³As a robustness check, we find similar results when classifying risks into systematic and idiosyncratic components; see Figure A.3.

state, the three realized R^2 series are very similar, which is key to our bias argument. All three series also vary with the realized market state, with the levels co-moving across realized scenarios. The paper’s claim does not depend on the sign of this aggregate slope; what matters is that the three series move together *within* each market-state bin, so the bear–base wedge in macro attention cannot be rationalized by a within-state difference in macro content of realized outcomes. Panel B shows the average ex ante macro attention that analysts assign to the bear, base, and bull cases.

Two findings stand out. First, the three attention series are relatively flat across market states compared with their realized R^2 benchmarks. We note that some flatness in the ex ante attention series is mechanical: macro attention is measured from reports written at t , before the forward 252-day return is observed, and annual market returns are weakly predictable, so under rational expectations the ex ante series should not closely track the bins on the horizontal axis. Realized CAPM R^2 , in contrast, is computed over the same 252-day window used for binning, so it covaries mechanically with the conditioning variable. The first finding is therefore a measurement asymmetry as much as a behavioral claim, and we lean on the bear–base gap within market state as the cleaner evidence. Second, and more central to the paper, macro attention in the bear case is consistently and substantially higher than in the base or bull case. This within-state wedge cancels any forward-looking attenuation common to all three scenarios, and it is the empirical signature of Macro Bear Bias: even when the realized macro component of risk is similar across scenarios conditional on market state, analysts continue to describe downside outcomes as disproportionately macro-driven.

Figure 7 makes this pattern especially clear by plotting the bear–base gap in macro attention together with the corresponding bear–base gap in realized R^2 across market states. The realized R^2 gap is close to zero when the subsequent annual market excess return is below 12%, and turns negative when the market excess return is above 12%. In contrast, the bear–base gap in macro attention remains significantly positive throughout. In other words, analysts consistently portray the bear case as much more macro-driven than the base case, even though the realized difference in

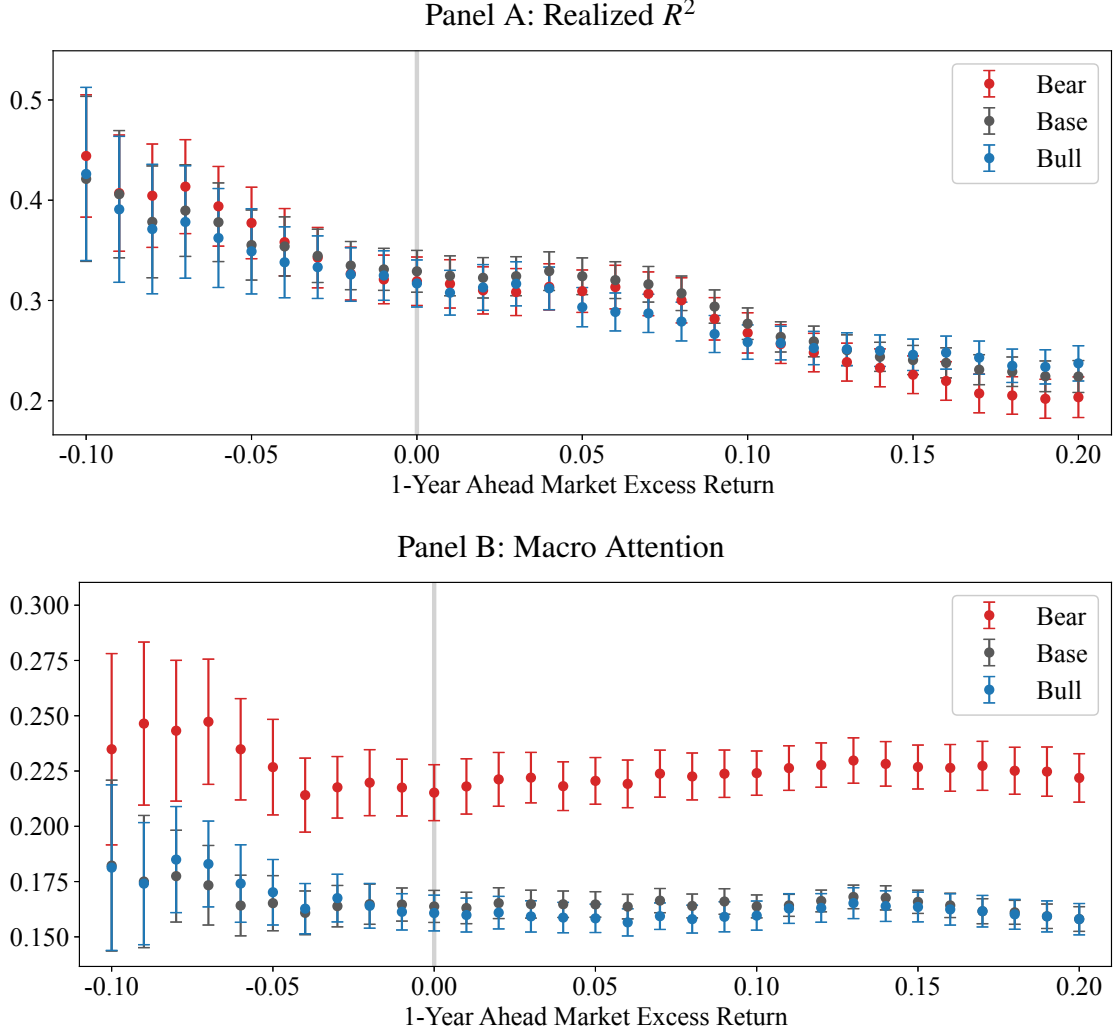


Figure 6 Realized R^2 and Macro Attention Conditional on Market States.

Note: This figure shows the realized R^2 and macro attention conditional on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention within each realized scenario (bear, base, and bull). Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear/base/bull cases. Error bars show 95% confidence intervals from constant-only regressions.

macro risk is economically small or even negative, depending on the market state. This divergence provides direct evidence of Macro Bear Bias.⁴

These patterns point to two possible explanations for Macro Bear Bias. The first is that analysts implicitly treat realized macro risk as roughly constant across market states, while at the same time

⁴Figure A.4 shows that the results are similar when risks are classified into systematic and idiosyncratic.

mechanically assigning extra weight to macro risk in bear cases. Under this explanation, analysts do condition on downside scenarios, but they fail to recognize that the importance of macro risk varies with the market environment. The second possibility is that analysts understand that realized R^2 declines with market returns, but still attach a persistently high probability to a severely negative future market outcome regardless of the prevailing market state. Under this interpretation, the elevated macro attention in bear cases would reflect a belief that downside scenarios are tightly linked to large aggregate market declines.

The evidence is more consistent with the first explanation. If the second explanation were correct, two implications should follow. First, base-case macro attention should track the slope of realized base-case R^2 across market states, in whichever direction Figure 6 establishes that slope. Second, bull-case macro attention should be materially lower than base-case macro attention, since the second explanation implies analysts assign less macro risk to more favorable scenarios. The data support neither implication. Base-case macro attention is largely flat across market states (Panel B of Figure 6), and bull-case macro attention is not materially below base-case macro attention—the bull–base gap in Table 1 is –0.40 percentage points, an order of magnitude smaller than the bear–base gap of 6.47. Viewed in concert, these findings suggest that analysts are not merely mapping different scenarios into different expected market states; rather, Macro Bear Bias reflects a systematic tendency to over-attribute downside outcomes to macro risks in bear-case narratives.

One interpretation of the first explanation is that the bias is state-independent, which points to two cognitive mechanisms. The first is the availability heuristic (Tversky and Kahneman, 1973). When analysts construct a bear case, they may rely on resemblance to a canonical macro-crisis prototype—a severe, overwhelmingly systematic downturn—rather than on a state-contingent assessment of systematic risk. Because prototype matching is unconditional, it naturally predicts the regime-independent pattern we document. The contrast with idiosyncratic framing follows directly: firm-specific disasters do not share a common prototype and therefore are less likely to shape bear-case narratives. The second mechanism is anchoring with insufficient adjustment

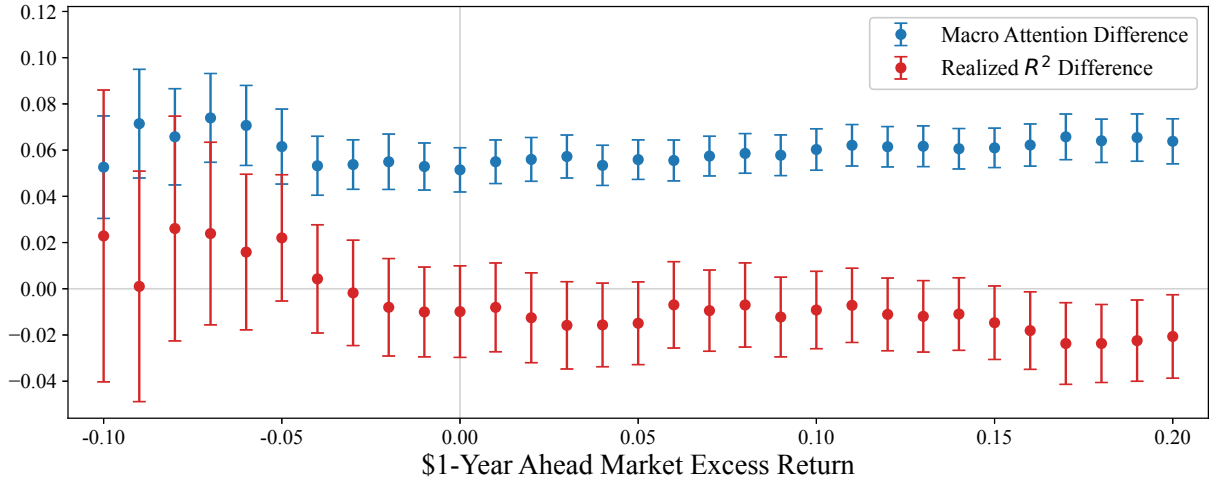


Figure 7 Bear – Base Gaps in Macro Attention and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bear–base differences in macro attention and realized CAPM R^2 on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention in the bear and base cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear or realized base cases. The plotted values represent the bear-minus-base differences. Error bars show 95% confidence intervals from constant-only regressions.

([Tversky and Kahneman, 1974](#)): even when firm-specific information calls for a different framing, analysts may adjust too little away from the initial macro template.

A natural objection is that the pattern could be mechanical: bear cases describe worse outcomes, which tend to co-occur with macro downturns, so bear narratives should naturally emphasize macro factors more. However, the regime-conditional analysis in Figures 6 and 7 directly addresses this. Conditional on the same forward market state, realized CAPM R^2 is similar across bear, base, and bull outcomes, yet the macro-attention gap remains large and positive. If elevated bear-case macro attention were a rational response to genuinely higher macro content in bear outcomes, the gap should shrink toward zero once we condition on market state—it does not. In unreported tests, we find no economically or statistically significant relationship between downside beta and MacroGap conditional on market state, further confirming that the pattern reflects misattribution rather than rational conditioning.

2.5 Bear-Case Macro Topics

The preceding evidence establishes that bear-case narratives systematically over-emphasize macro risk relative to the realized ex post benchmark. We now characterize the *content* of this emphasis: which macroeconomic topics populate bear narratives, and do analysts update their topic-level attention in response to news or anchor to persistent templates?

To identify topic-level narrative content, we apply a second LLM extraction step that classifies each macro-risk mention extracted from the bear-case narrative into one of 52 macro topic categories drawn from [Zhou \(2025\)](#), excluding 7 corporate finance topics from their original 59-topic taxonomy. A single bear-case narrative may therefore contribute multiple topic labels, one per mention. Appendix [A.2](#) describes the prompt and Table [A.1](#) reports each topic’s description and summary statistics. Following [Zhou \(2025\)](#), bear-case analyst attention to a topic in a given month is the share of mentions associated with that topic across all bear-case narratives released in that month. We benchmark this against Wall Street Journal (WSJ) coverage. WSJ attention to a topic is the monthly share of WSJ articles tagged with that topic. The two series therefore use different denominators—mention shares on the analyst side and article shares on the WSJ side—so the comparison in Figure [8](#) and Table [3](#) is best read as comovement in standardized time-series variation rather than a level-for-level match.

Table [3](#) reports average topic-level attention and time-series co-movement with the WSJ. Panel A ranks topics by average monthly bear-case analyst attention. *Economic Growth and Recovery Outlook* is the dominant bear-narrative topic, accounting for 10.66% of bear-case attention on average—more than four times its WSJ share of 2.36% (WSJ rank: 12). This disproportion suggests that macroeconomic growth narratives function as a canonical bear-case template: analysts invoke them persistently, at far higher rates than contemporaneous news coverage would predict. *COVID-19 Pandemic and Vaccine Developments* ranks second at 5.36%. *Market Activity and Financial Performance* ranks third (4.75%), despite being the single most-covered topic in the WSJ (rank 1, 5.44%), suggesting analysts allocate less narrative emphasis to general market conditions than the news cycle does. Inflation (*Consumer Price Index and Inflation Trends*, rank 4) and trade

Table 3 Bear Analyst Attention and WSJ Attention Across Key Topics

Note: This table reports macro topics in analysts’ bear-case narratives and corresponding attention to the same topics in Wall Street Journal articles. Bear analyst attention for a given topic is measured as the share of topic mentions in the bear-case narrative associated with this topic across analyst reports. WSJ attention for a given topic is measured as the average monthly share of Wall Street Journal articles associated with that topic over the sample period. Panel A shows the top five topics ranked by average monthly bear analyst attention. Panel B shows the top five topics with the highest correlation between monthly bear analyst attention and monthly WSJ attention. Bear Rank and WSJ Rank report the rankings of each topic by average attention within analyst bear narratives and the Wall Street Journal, respectively.

Panel A: Top Five Topics by Average Monthly Bear Analyst Attention

Topic	Bear Avg.	Bear Rank	WSJ Avg.	WSJ Rank
Economic Growth and Recovery Outlook	10.66%	1	2.36%	12
COVID-19 Pandemic and Vaccine Developments	5.36%	2	5.34%	2
Market Activity and Financial Performance	4.75%	3	5.44%	1
Consumer Price Index and Inflation Trends	2.40%	4	0.86%	27
International Trade Relations and Economic Policies	2.31%	5	1.60%	16

Panel B: Top Five Topics by Correlation Between Bear Analyst Attention and WSJ Attention

Topic	Corr.	Bear Avg.	Bear Rank	WSJ Avg.	WSJ Rank
COVID-19 Pandemic and Vaccine Developments	0.76	5.36%	2	5.34%	2
Economic Growth and Recovery Outlook	0.49	10.66%	1	2.36%	12
Consumer Behavior Amid Economic Strains	0.38	1.91%	7	0.24%	49
Geopolitical Tensions and Economic Impacts	0.38	0.46%	20	4.40%	3
Economic Stimulus and Government Interventions	0.37	0.40%	25	0.46%	40

policy (*International Trade Relations and Economic Policies*, rank 5) round out the top five, each receiving materially more analyst attention than their WSJ ranks imply. The reverse asymmetry is equally informative. *Geopolitical Tensions and Economic Impacts* is the third most-covered WSJ topic (4.40% share) but ranks only 20th in bear-case attention (0.46%)—roughly a tenfold under-weight relative to its media prominence. If the availability heuristic operated through generic media exposure, geopolitics should occupy a much larger share of bear narratives. Instead, analysts over-weight a narrow band of recession-and-growth topics and under-weight geopolitical risk, suggesting that the operative version of availability is not “whatever dominates the news cycle” but “the canonical macro-crisis template,” centered on growth, inflation, and recession risk. This

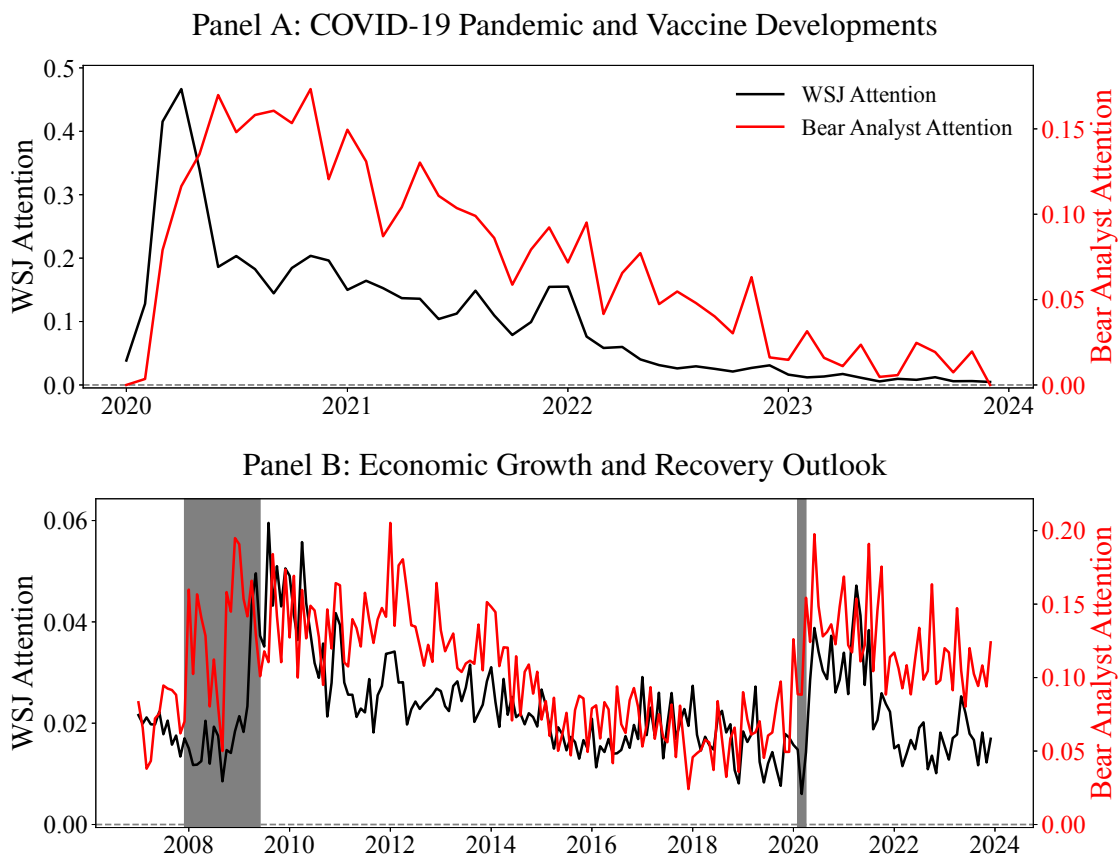


Figure 8 Bear Analyst Attention and WSJ Attention Over Time for Selected Topics.

Note: This figure plots monthly bear analyst attention (red, right axis) alongside monthly Wall Street Journal (WSJ) attention (black, left axis) for two representative macro topics. Bear analyst attention for a given month is the share of bear-case narrative mentions across all reports in that month associated with the topic. WSJ attention is the monthly share of WSJ articles associated with the topic. Panel A covers *COVID-19 Pandemic and Vaccine Developments*. Panel B covers *Economic Growth and Recovery Outlook*; gray bands indicate the Global Financial Crisis recession period (2008-2009) and the COVID-19 recession period (2020).

selectivity foreshadows the post-GFC $\times$ high-systematic interaction and the GFC-at-MS result in Section 4.1.

Panel B ranks topics by the time-series correlation between monthly bear-case analyst attention and WSJ attention. COVID-19 leads (0.76), followed by economic growth (0.49). High correlations confirm that analysts respond to the news environment in their narrative topic choices. The critical question, however, is whether that response is symmetric: do analysts reduce topic emphasis as quickly as they raise it, or do bear-case templates persist long after the originating events have receded?

Figure 8 examines this asymmetry directly. Each panel overlays monthly bear-case analyst attention (red, right axis) against WSJ attention (black, left axis) for the two highest-correlation topics. Panel A, covering COVID-19, shows a sharp joint spike in early 2020. WSJ attention subsequently declines steeply, reaching near zero by 2023; bear-case analyst attention remains materially elevated for roughly two additional years. Panel B, covering economic growth, replicates this pattern across both the Global Financial Crisis and the COVID-19 period: analyst attention rises alongside WSJ coverage during each crisis but reverts far more slowly, remaining elevated long after media attention has subsided.

The asymmetric decay is consistent with template-based updating driven by the availability heuristic. If analysts anchor bear narratives to salient crisis templates—deploying them on the basis of memorability rather than current relevance—those templates should persist in narratives well after the originating events recede. The natural alternative is a *news-tracking* benchmark, under which bear-case analyst attention and WSJ attention move in parallel: topics enter and exit narratives proportionally to their presence in the news environment. The data reject this news-tracking benchmark, with bear-case attention persisting long after WSJ coverage subsides. We caution, however, that the news-tracking benchmark is not the same as a fully rational benchmark. WSJ coverage tracks editorial novelty and reader demand; the economically relevant risk attached to a macro topic—post-GFC credit impairment, post-COVID supply-chain disruption—can decay much more slowly than the news cycle. A rational analyst weighting topics by current risk-relevance rather than by media salience would therefore look like our “availability” pattern. The asymmetric-decay evidence alone cannot separate availability bias from sluggish-but-rational updating on persistent risks; we rely on the cross-sectional tests in Section 4.1—which condition on observable risk exposure—to discriminate between the two. We develop and test this mechanism formally in Section 4.1.

3 Conceptual Framework and Forecast Error Evidence

We now formalize the bias documented in Section 2. Analysts routinely describe bear, base, and bull cases to explain the range of possible future outcomes. In principle, these narratives should reflect how different sources of risk contribute to firm value conditional on the prevailing market environment. In practice, however, analysts appear to assign disproportionately high macro emphasis to downside scenarios, even when the conditional macro component of risk is not especially large.

A central implication of this framework is that the bias does not remain confined to the bear narrative itself. Because scenario narratives shape analysts' overall valuation judgments, excessive macro emphasis in the bear case can spill over into the analyst's base-case forecast, making it too pessimistic relative to subsequently realized outcomes. This link connects Macro Bear Bias to the prediction of analyst forecast errors.

3.1 Benchmark: Conditional Attribution of Risk

Consider firm i 's future return:

$$R_{i,t+1}^{\text{realized}} = \alpha_{i,t} + \beta_i M_{t+1} + \varepsilon_{i,t+1}, \quad (1)$$

where M_{t+1} denotes the future market excess return, β_i measures the firm's exposure to aggregate macro conditions, and $\varepsilon_{i,t+1}$ captures firm-specific shocks. The key feature of the benchmark is that the importance of macro risk is conditional: it should be evaluated relative to the market state prevailing when the analyst forms the forecast.

Let s_t denote the information available at time t about the market environment. When an analyst constructs scenario $k \in \{\text{bear, base, bull}\}$, the rational benchmark requires that the macro emphasis assigned to that scenario reflect the analyst's expectation of how important macro forces would be *if scenario k were to materialize*. A bear narrative should emphasize macro risk to the extent that bear-case outcomes are genuinely macro-driven; a bull narrative should do so to the

extent that bull-case outcomes are. Empirically, we evaluate this benchmark by classifying reports ex post— $k = \text{bear}$ if the firm’s stock price ultimately falls below the analyst’s bear-case target, $k = \text{base}$ if it falls between the bear and bull targets, and $k = \text{bull}$ if it exceeds the bull target—and comparing the ex ante attention to the ex post macro content of each realized outcome. Formally, let

$$\text{Macro Importance}_{i,t+1} \equiv \frac{\beta_i^2 \text{Var}(M_{t+1} \mid s_t, k)}{\text{Var}(R_{i,t+1} \mid s_t, k)}$$

denote the fraction of firm-level return variance attributable to market movements, conditional on the prevailing market state s_t and the realized scenario k . Here β_i is the within-cell population regression coefficient of $R_{i,t+1}$ on M_{t+1} , so that $\varepsilon_{i,t+1}$ is orthogonal to M_{t+1} within the conditioning cell and the ratio is a clean variance share.⁵ The empirical CAPM R^2 computed in Section 2.4 is the sample analogue of this conditional object within the corresponding market-state and realized-scenario cell. The rational benchmark is:

$$A_{i,k,t} = g(\mathbb{E}_t[\text{Macro Importance}_{i,t+1} \mid s_t, k]), \quad (2)$$

where $g(\cdot)$ is an increasing mapping from the underlying importance of macro risk into narrative emphasis. Under this benchmark, two properties should hold. First, macro attention should vary with the prevailing market environment, since the contribution of macro forces to firm outcomes is state-dependent. Second, differences in macro attention across scenarios should reflect genuine differences in the conditional relevance of macro risk, not a mechanical template that treats bad outcomes as macro-driven. The bear case may legitimately receive more macro emphasis, but only when conditional macro exposure truly justifies it.

This benchmark provides the reference for interpreting the data: if analysts are well calibrated, ex ante macro attention should track the ex post macro-related component of realized variation conditional on market state. Divergence between the two signals misattribution.

⁵Without the within-cell orthogonality, the ratio would also include a $2\beta_i \text{Cov}(M, \varepsilon \mid s_t, k)$ term, so the conditioning is what makes the object a variance share rather than a covariance-laden index.

3.2 Macro Bear Bias: Over-attributing Macro Risks in Bear Narratives

Macro Bear Bias arises when analysts depart from this benchmark by assigning excessive macro weight specifically to the bear narrative. Formally, let observed macro attention satisfy

$$A_{i,k,t} = A_{i,k,t}^* + b_{i,t} \cdot \mathbf{1}\{k = \text{bear}\}, \quad b_{i,t} \geq 0, \quad (3)$$

where attention shares lie in $[0, 100]$, so the inequality $0 \leq b_{i,t} \leq 100 - A_{i,\text{bear},t}^*$ ensures feasibility and the bear-case excess is offset by a corresponding reduction in non-macro attention.⁶ The parameter $b_{i,t}$ captures the analyst's excess tendency to explain downside scenarios using aggregate macro forces, beyond what is justified by the conditional importance of macro risk. We allow $b_{i,t}$ to vary across firms and over time, and the empirical hypothesis is that $\mathbb{E}[b_{i,t}] > 0$ in the relevant population—a unit with $b_{i,t} = 0$ is permitted, but the average bear-case excess is strictly positive.

Under this formulation, the distortion is not that analysts discuss macro risk per se. Rather, it is that they over-explain downside outcomes through macro forces. The bear narrative becomes disproportionately macro-heavy even when, conditional on the same market state, the realized macro component of outcomes is similar to that in the base or bull case. This is why Macro Bear Bias is best understood as a misattribution bias.

The empirical counterpart of this bias is the bear–base macro-attention gap:

$$\text{MacroGap}_{i,t} = \text{MacroAttention}_{i,t}^{\text{bear}} - \text{MacroAttention}_{i,t}^{\text{base}}. \quad (4)$$

A larger $\text{MacroGap}_{i,t}$ indicates that the analyst places substantially greater macro emphasis on the downside scenario than on the central scenario. Under the benchmark, this gap should reflect genuine conditional differences in macro relevance. Under Macro Bear Bias, however, the gap is too large because the bear narrative receives an additional, bias-driven macro overlay.

⁶For interior values of $A_{i,\text{bear},t}^*$ the upper bound is slack; the constraint binds only when the unbiased bear-case macro share is already near the ceiling.

3.3 Economic Intuition

The framework distinguishes between two objects: the true conditional importance of macro risk for firm outcomes, and the analyst’s narrative attribution of macro risk across scenarios. If analysts are well calibrated, these should move together. Macro Bear Bias breaks this alignment. Analysts apply a coarser rule: downside scenarios are treated as macro-driven almost by construction, regardless of the prevailing market state. The distortion is not that analysts mismeasure macro risk on average but that they do so *asymmetrically*, loading macro explanations disproportionately into bear narratives.

This asymmetry matters because scenario narratives are not merely descriptive—they organize analysts’ thinking about valuation. A macro-heavy bear case can contaminate the base-case forecast itself, making it too pessimistic relative to subsequently realized outcomes. We develop microfoundations for this spillover through two channels and derive a cross-sectional prediction linking both channels to the direction of analyst conviction.

The primary channel operates through discount rates. A bear case described as predominantly macro-driven leads the analyst to perceive elevated systematic risk, implicitly adopting a higher equity risk premium when discounting base-case cash flows. Let the base-case price target be $P^{\text{base}} = \sum_{\tau} CF_{\tau} / (1 + r_{\tau})^{\tau}$, where $r_{\tau} = r_f + \tilde{\beta}_i \cdot \text{ERP}$. Macro Bear Bias inflates the analyst’s perceived beta *multiplicatively*, so that the distortion scales with the firm’s true systematic exposure:

$$\tilde{\beta}_{i,t} = \beta_i (1 + \phi \cdot \text{MacroGap}_{i,t}), \quad \phi > 0. \quad (5)$$

Under this specification, the incremental discount-rate wedge is $\beta_i \phi \cdot \text{MacroGap}_{i,t} \cdot \text{ERP}$: a high-beta firm is hit by a larger absolute wedge than a low-beta firm at the same MacroGap. The wedge raises the discount rate and depresses P^{base} even if point estimates of base-case cash flows are unchanged, and the scaling by β_i generates the $\beta_i \times \text{MacroGap}_{i,t}$ interaction tested in the empirical specifications

below.⁷ This microfoundation maps directly onto the reduced-form spillover parameter θ in Eq. (9): θ captures the product of ϕ and the sensitivity of the price target to the discount rate.⁸

A numerical example anchors the economic magnitude. At the sample-mean values $\hat{\beta}_{\text{CAPM}} = 1.13$ (Table 2), a six-percent equity risk premium, and a bias coefficient $\phi = 0.005$, a one-standard-deviation MacroGap shift of about 6.3 percentage points raises the perceived beta from 1.13 to $1.13 \times (1 + 0.005 \times 6.3) = 1.166$. The incremental discount-rate wedge is therefore $\hat{\beta}_{\text{CAPM}} \phi \text{ MacroGap} \cdot \text{ERP} = 1.13 \times 0.005 \times 6.3 \times 6\% \approx 21 \text{ bp}$. At an effective valuation duration of ten years, modified duration is $10/(1+r) \approx 9.4$, so the implied compression of the base-case target price is $-9.4 \times 21 \text{ bp} \approx -2.0\%$. This is comparable to the median 1–2% standardized base-case forecast error documented in Table 4, suggesting the discount-rate channel can plausibly carry the magnitude of the observed bias.

The same calculation under $\beta = 2.0$ (high-beta firm) yields a wedge of $2.0 \times 0.005 \times 6.3 \times 6\% \approx 38 \text{ bp}$ and a target-price compression of about -3.5% ; under $\beta = 0.5$ (low-beta firm) the wedge is about 9 bp and the compression is about -0.9% . The wedge therefore scales roughly four-to-one between extreme high- and low-beta firms in the sample, which delivers the cross-sectional content of the $\beta_i \times \text{MacroGap}_{i,t}$ interaction in Eq. (10): high-beta firms should display the largest forecast-error response to MacroGap.

A complementary channel operates through narrative coherence. Analysts strive for internal consistency across scenarios: a bear case that emphasizes macro headwinds pulls the base-case narrative toward acknowledging those same headwinds, even when the base-case probability assessment has not changed. This cognitive-consistency pressure shifts the base-case price target downward independently of the discount rate, and helps explain why Macro Bear Bias predicts forecast errors even after controlling for β_i .

Both channels predict that the *direction* of an analyst's base-case conviction relative to the

⁷An additive specification, $\tilde{\beta}_i = \beta_i + \phi \cdot \text{MacroGap}_{i,t}$, would deliver a wedge $\phi \cdot \text{MacroGap}_{i,t} \cdot \text{ERP}$ that is constant in β_i and would therefore *not* predict the interaction. The multiplicative form is the right microfoundation for the empirical specification.

⁸Some DCF implementations weight scenarios explicitly: $P^{\text{reported}} = w_{\text{bear}} P^{\text{bear}} + w_{\text{base}} P^{\text{base}} + w_{\text{bull}} P^{\text{bull}}$. If availability inflates w_{bear} or macro narratives compress P^{bear} , the composite target shifts down. This is a reduced-form version of the discount-rate channel.

consensus should predict the structure of the bear-case narrative. An optimistic analyst—whose base-case price target exceeds the consensus—needs a bear case sufficiently distant from the elevated base case. Firm-specific risks (product delays, margin compression) may not generate enough downside distance, because they are bounded by the firm’s own cash-flow distribution. A macro bear narrative, by contrast, can invoke a market-wide downturn that depresses the stock regardless of firm fundamentals, producing the requisite spread. The discount-rate channel amplifies this logic: the optimistic analyst signals confidence in firm-specific fundamentals, so the residual explanation for downside must be systematic. Conversely, an analyst whose base case is below consensus has already embedded pessimism into the central scenario; firm-specific downside narratives suffice, and the macro alibi is less needed. This reasoning predicts that base-case position relative to consensus should covary with the macro content of the bear case: above-consensus base targets associated with larger MacroGap, below-consensus base targets with smaller MacroGap. We test this prediction in Section 4.1.2; the data support it but in an asymmetric form—the cross-section is driven primarily by the below-consensus margin, not the above-consensus margin.

These microfoundations also clarify why comparing ex ante macro attention with ex post realized R^2 is informative, and why a linear specification understates the predictive content of Macro Bear Bias. As documented in Section 2.4, bear-case macro attention is essentially flat across market environments. But the *consequences* of this flat gap are state-dependent: in strong markets, the same macro-heavy bear narrative diverges more sharply from realized conditions, generating larger forecast errors; in weak markets, the template is closer to correct, and the distortion is smaller. When macro attention in bear narratives remains elevated even though conditional realized R^2 is similar across scenarios, that wedge is direct evidence of Macro Bear Bias. A linear specification, by construction, recovers only the variance-weighted average of these state-specific effects: it is the right summary if the effects share a sign, but it understates predictive content whenever the effect is positive in one regime and negative or zero in another and the cancellation is large. We do not claim that state dependence *must* compress the slope to zero; we claim that the linear average is the wrong summary of a state-dependent mechanism. The nonlinear specification in Section 3.6 is

designed to capture this interaction, and it explains why out-of-sample R^2 improves dramatically once the model allows the effect of narrative attention to vary with market conditions.

3.4 Forecast Error Prediction

The key implication of the framework is that Macro Bear Bias should predict subsequent errors in analysts' base-case forecasts. Let the realized return and the analyst's base-case forecasted return be

$$R_{i,t+1}^{\text{realized}} \equiv \frac{P_{i,t+1}^{\text{realized}}}{P_t}, \quad R_{i,t+1}^{\text{base}} \equiv \frac{P_{i,t+1}^{\text{base}}}{P_t}, \quad (6)$$

and define the base-case forecast error as

$$\text{Forecast Error}_{i,t+1} = R_{i,t+1}^{\text{realized}} - R_{i,t+1}^{\text{base}}. \quad (7)$$

Under this definition, a positive forecast error indicates that the analyst's base-case forecast is too pessimistic relative to the subsequently realized return.

Absent bias, the analyst's base-case forecast would reflect expected market conditions:

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t, \quad (8)$$

where $\mu_t = E_t[M_{t+1}]$ is the analyst's expectation of the future market excess return. Macro Bear Bias enters when excessive macro emphasis in the bear narrative spills over into the base-case forecast. We capture this spillover through a reduced-form specification: the base-case forecast is shifted downward in proportion to both the firm's systematic exposure β_i —because the bias operates through the macro channel—and the size of the bear-base attention gap. The parameter $\theta > 0$ is a reduced-form coefficient. The sign prediction in Eq. (11) holds for any spillover function that is monotonically increasing in MacroGap; we adopt the linear form below for analytical transparency,

but the empirical content is the sign of the derivative, not the specific functional form.

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t - \theta \beta_i \text{MacroGap}_{i,t}, \quad \theta > 0. \quad (9)$$

This specification formalizes the idea that stronger Macro Bear Bias makes the analyst's central forecast too low. The distortion operates through the macro channel: when analysts over-attribute downside outcomes to aggregate macro forces, they embed too much downside macro risk into the base case.

Substituting the realized-return process in Eq. (1) and the biased base-case forecast in Eq. (9) into the forecast-error definition in Eq. (7) yields

$$\text{Forecast Error}_{i,t+1} = \beta_i (M_{t+1} - \mu_t) + \theta \beta_i \text{MacroGap}_{i,t} + \varepsilon_{i,t+1}. \quad (10)$$

The intercept $\alpha_{i,t}$ enters both Eq. (1) and Eq. (9) identically and therefore cancels in the forecast error. This equation generates a sharp empirical prediction. Holding fixed the realized market shock and other determinants of firm outcomes, a larger bear-base macro-attention gap should predict a larger, more positive forecast error. Equation (10) implies

$$\frac{\partial \text{Forecast Error}_{i,t+1}}{\partial \text{MacroGap}_{i,t}} = \theta \beta_i, \quad (11)$$

which is positive for firms with $\beta_i > 0$, zero for zero-beta firms, and negative for negative-beta firms. The empirical sample is large-cap analyst-covered equities with $\hat{\beta}_{\text{CAPM}}$ centered at 1.13 (Table 2); the positive-beta case is therefore the empirically dominant one, and we state the testable prediction as $\theta \beta_i > 0$ for $\beta_i > 0$.

The intuition is straightforward. A larger $\text{MacroGap}_{i,t}$ indicates that the analyst places unusually strong macro emphasis on the downside scenario relative to the base case. If this reflects Macro Bear Bias rather than a rational assessment of conditional macro risk, then the analyst sets the base-case forecast too pessimistically. When outcomes are later realized, the forecast tends to be too low, producing a positive forecast error.

We now test this prediction empirically. Equation (10) motivates a panel regression of forecast errors on MacroGap with firm fixed effects and a market-state control. Because $\alpha_{i,t}$ has already cancelled in the derivation of Eq. (10), the role of firm fixed effects is to absorb any residual persistent firm-level forecast-error component (for example, time-invariant analyst optimism on a given name) that is not modeled in the framework but could correlate with MacroGap.⁹ Because β_i enters the slope multiplicatively rather than additively, firm fixed effects do not remove cross-sectional variation in β_i ; the pooled OLS slope on MacroGap instead recovers a variance-weighted average of the firm-specific products $\theta\beta_i$. The forward-looking market surprise $\beta_i(M_{t+1} - \mu_t)$ enters the residual, so identification requires that MacroGap does not predict future market returns. We examine this prediction in two steps. Section 3.5 uses a linear panel regression; Section 3.6 uses a flexible nonlinear specification that allows the effect to vary across market states.

3.5 Forecast Error Prediction: Empirical Evidence

Guided by the framework in Section 3.4, our central empirical prediction is that stronger Macro Bear Bias should be associated with more positive subsequent base-case forecast errors. Intuitively, when analysts place excessive macro emphasis on downside scenarios, they embed too much downside macro risk into the central forecast. As a result, the base-case forecast becomes too pessimistic relative to subsequently realized returns.

To test this prediction, we estimate the following specification:

$$\text{Forecast Error}_{i,t+1} = a + b_1 \text{MacroGap}_{i,t} + b_2 M_{i,t} + \Gamma' X_{i,t} + \gamma_i + u_{i,t+1}, \quad (12)$$

where $\text{MacroGap}_{i,t}$ is the bear–base gap in macro attention, $M_{i,t}$ is the past 252-day market excess return prior to report release, $X_{i,t}$ is a vector of firm-level controls, and γ_i denotes firm fixed effects. In the baseline specification, $X_{i,t}$ contains only the stock’s own past 252-day return before report release. We include $M_{i,t}$ to control for the market state prevailing when the analyst issues the report.

⁹If $\alpha_{i,t}$ is interpreted as time-varying, only its time-invariant component α_i is absorbed by firm fixed effects; the within-firm time-varying residual $\alpha_{i,t} - \alpha_i$ enters the regression residual and identification additionally requires it to be uncorrelated with MacroGap. Equation (10)’s derivation does not require this assumption because α cancels there; we flag it here only because the empirical specification could in principle face it.

The framework predicts $b_1 > 0$: firms with stronger Macro Bear Bias should exhibit larger positive subsequent forecast errors.

For the empirical analysis, we go beyond the bear–base macro-attention gap and also examine the bull–base macro-attention gap, as well as the corresponding gaps in firm-specific attention as placebo tests. This design helps distinguish Macro Bear Bias from more general differences in narrative intensity across scenarios. If forecast errors are specifically driven by Macro Bear Bias, then the relevant predictor should be the bear–base gap in macro attention, rather than scenario differences in upside narratives or in firm-level risk attention.

The regression results are reported in Table 4. Across columns (1), (5), and (6), the coefficient on the bear–base macro-attention gap is positive and statistically significant, with t -statistics of 2.37, 2.07, and 3.25, respectively. These results indicate that firms for which analysts place relatively greater macro emphasis in the bear case than in the base case subsequently exhibit larger base-case forecast errors. Under our definition of forecast error, this means that analysts set base-case forecasts that are too low for these firms. The evidence is therefore consistent with Macro Bear Bias distorting analysts’ base-case expectations in a systematically pessimistic direction.

By contrast, the bull–base macro-attention gap is not statistically significant, which is consistent with the view that the bias is specific to downside macro narratives rather than to scenario differentiation more broadly. The placebo tests based on firm-specific attention provide weaker and less consistent evidence. On the whole, the results suggest that the predictive content comes primarily from the downside macro component of analyst narratives, in line with the mechanism of Macro Bear Bias.

The univariate-to-multivariate comparison across Table 4 is itself informative. In the univariate specification of column (1), the Macro Bear–Base coefficient is 0.0012 ($t = 2.37$); once all four attention gaps enter jointly in column (6), it nearly doubles to 0.0023 ($t = 3.25$). The pattern is a textbook suppression effect: because the macro and firm bear–base gaps are mechanically negatively correlated—an analyst who allocates more bear-case attention to macro factors necessarily allocates less to firm-specific factors given the compositional constraint $A^{\text{macro}} + A^{\text{firm}} + A^{\text{industry}} \leq 100$ —the univariate Macro Bear–Base coefficient is attenuated toward zero by the omitted Firm Bear–Base

gap. Column (6) recovers the partial effect by netting out this within-bear-case substitution. We read column (6) as the better-specified linear estimate of the bias coefficient, and we use its magnitude (~ 0.23 basis points of one-year-ahead forecast error per percentage-point of MacroGap, or roughly 1.5% for a one-standard-deviation move in MacroGap) as the relevant benchmark for the nonlinear evidence in Section 3.6. The motivation for the RFF specification is therefore not that the linear effect is null—column (6) shows it is not—but that even the properly partialled linear effect is an order of magnitude smaller than the state-dependent effect the nonlinear model recovers.

A second feature of column (6) deserves comment. The Firm Bear–Base coefficient is 0.0018 ($t = 2.28$)—significant at the 5% level—and within striking distance of the Macro Bear–Base coefficient of 0.0023. A clean macro-specific story would predict a null for the firm placebo, and the data fall short of that benchmark.¹⁰ We read the placebo result in two ways. First, the placebo’s statistical significance is consistent with a broader phenomenon—any cross-scenario differentiation in attention may signal analyst miscalibration, regardless of the dimension along which the differentiation occurs. Second, the macro coefficient is nonetheless larger in magnitude, and the bull–base macro and bull–base firm placebos remain null. The substantive interpretation we carry forward is that Macro Bear Bias is the predominant but not exclusive channel through which scenario-differentiation predicts forecast errors, and we soften the corresponding statement in the conclusion accordingly.

¹⁰A Wald test of equality between the Macro Bear–Base and Firm Bear–Base coefficients in column (6) would directly address whether the data distinguish a macro-specific channel from a more general scenario-differentiation channel. We do not estimate the Wald statistic in the current draft and add this to the planned robustness pass.

Table 4 Forecast Error Regressions

Note: This table reports empirical results for forecast error prediction regression (12). Columns (1)–(4) include each gap individually. Column (5) includes the two macro gaps jointly. Column (6) includes all four narrative gap variables simultaneously. Firm fixed effects are included in all specifications. Standard errors are clustered by firm and month. t-statistics are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Attention Gaps						
Macro Bear – Base	0.0012** (2.37)				0.0011** (2.07)	0.0023*** (3.25)
Macro Bull – Base		0.0009 (1.03)			0.0005 (0.53)	0.0002 (0.23)
Firm Bear – Base			0.0006 (1.10)			0.0018** (2.28)
Firm Bull – Base				-0.0002 (-0.25)		-0.0004 (-0.57)
Control Variables						
Market Excess Return	-0.1467 (-1.56)	-0.1477 (-1.57)	-0.1474 (-1.57)	-0.1473 (-1.57)	-0.1469 (-1.56)	-0.1470 (-1.57)
Firm Excess Return	-2.3962 (-0.15)	-2.3370 (-0.14)	-2.1720 (-0.13)	-2.3007 (-0.14)	-2.4154 (-0.15)	-2.1880 (-0.13)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	43,313	43,313	43,313	43,313	43,313	43,313
Within R^2	0.0004	0.0003	0.0003	0.0003	0.0004	0.0006

3.6 Nonlinear Forecast Error Prediction and Portfolio Performance

Table 4 shows that the linear specifications have limited explanatory power: the within R^2 ranges only from 0.03% to 0.06%, even though the bear–base macro-attention gap significantly predicts subsequent forecast errors. This combination of statistical significance and low explanatory power suggests that Macro Bear Bias contains predictive information, but that its effect is not well summarized by a simple additive linear specification. In particular, a single slope coefficient may average over heterogeneous effects that depend on the prevailing market environment.

This interpretation follows naturally from the conceptual framework. Macro Bear Bias reflects a failure to condition appropriately on aggregate market states. The same bear–base macro-attention gap may therefore imply different forecast distortions depending on whether market conditions

are weak or strong. In weak market states, a given amount of downside macro emphasis may translate into one degree of pessimism in the base-case forecast; in stronger market states, the same narrative gap may imply a different distortion. More generally, the effect of Macro Bear Bias is likely governed by nonlinear interactions between narrative-based attention measures and market conditions, rather than by their separate linear effects.

To capture these nonlinearities, we follow [Kelly, Malamud, and Zhou \(2024\)](#) and use the random Fourier features (RFF) approach of [Rahimi and Recht \(2007, 2008\)](#). The raw predictors are the macro bear–base gap, macro bull–base gap, firm bear–base gap, firm bull–base gap, and the cumulative market excess return over the 252 trading days prior to the report release. We exclude lagged firm-level excess returns so that firm-specific information enters only through the attention measures extracted from analyst reports. Because the forecast error is defined as the difference between the realized one-year-ahead return and the analyst’s base-case forecast return, we use only training observations released at least one year before the test report. This timing convention avoids using realized returns that would not have been observable at the forecast date. For example, to predict the forecast error for a report released on June 1, 2021, the training sample consists of reports released before June 1, 2020.

Let G_t denote the 5×1 predictor vector for a given firm-date observation. The RFF procedure maps G_t into nonlinear features,

$$S_{i,t} = [\sin(\gamma\omega_i'G_t), \cos(\gamma\omega_i'G_t)]', \quad \omega_i \sim i.i.d. N(0, I), \quad (13)$$

where each ω_i defines a random projection of the original predictors, which is then transformed using sine and cosine functions. This transformation generates a rich set of nonlinear combinations of attention gaps and market states. As emphasized by [Kelly, Malamud, and Zhou \(2024\)](#), RFF can be interpreted as a wide two-layer neural network with random first-layer weights and estimated second-layer coefficients. Its advantage in our setting is that it introduces flexible nonlinearities while remaining computationally tractable and straightforward to evaluate out of sample. The

model size parameter P counts random projections ω_i ; because each projection contributes one sine and one cosine regressor, the ridge design matrix at model size P has $2P$ columns.

Following Kelly, Malamud, and Zhou (2024), we use a *rolling* training window of 60 observations: at each forecast date, the most recent 60 reports for the firm released at least one year before the test date enter the training sample, and older observations roll out. We set the bandwidth parameter to $\gamma = 2$ and generate up to $P = 180,000$ random projections (so up to 360,000 sine-cosine features) per firm. For each firm, we estimate out-of-sample forecasts at each rolling window over a grid of model sizes $P \in \{6, \dots, 180,000\}$ using a fixed ridge penalty $z = 1,000$. Each firm is forecasted separately using only its own attention-gap variables and the aggregate market state.

We next examine whether correcting for Macro Bear Bias improves the cross-sectional ranking of stocks. We evaluate portfolio performance using three sorting signals. The first is the nonlinearity-adjusted forecast, defined as the analyst’s base-case forecast return plus the out-of-sample forecast error predicted by the RFF model. This adjusted measure corrects the analyst’s base-case forecast for the predictable bias component associated with Macro Bear Bias. The second signal is the analyst’s raw base-case forecast return. The third signal is the RFF-predicted forecast error alone.

For each signal, we form long-short portfolios using an annual rebalancing strategy implemented separately at the end of each calendar month. Specifically, for each calendar month, we construct a portfolio that is rebalanced annually in that month, yielding 12 distinct annually rebalanced strategies. To examine performance when the sorting signal is strongest, we form decile-style long-short portfolios by going long the top 10% of stocks and short the bottom 10%. We construct both equal-weighted (EW) and value-weighted (VW) versions of each portfolio. We then average performance across the 12 calendar-month implementations and report the annualized Sharpe ratio, average monthly return, and CAPM alpha.

Table 5 shows a sharp contrast across the three signals. The Base Target signal, which sorts on the analyst’s base-case forecast return alone, generates no abnormal return. The VW long-short portfolio earns an average monthly return of -0.10% ($t = -0.18$) and a CAPM alpha of -0.30% ($t = -0.61$). The EW results are similarly flat, with an average return of -0.03% ($t = -0.05$).

Table 5 Portfolio Performance

Note: This table reports monthly long-short portfolio performance for three sorting signals. Each portfolio goes long the top decile and short the bottom decile. *Nonlinearity-adjusted Forecast* sorts on the analyst's base-case forecast return augmented with the RFF-predicted forecast-error correction. *Base Target* sorts on the analyst's raw base-case forecast return and serves as the benchmark. *Predicted Forecast Error* sorts on the RFF-predicted correction alone. All specifications use bandwidth $\gamma = 2$, shrinkage $z = 1,000$, and complexity $c = 3,000$. Portfolios are rebalanced annually across 12 calendar-month implementations. Annual SR is the annualized Sharpe ratio. Avg Return is the average monthly return of the long-short portfolio. CAPM α is the intercept from a CAPM time-series regression. VW and EW denote value-weighted and equal-weighted portfolios, respectively. t -statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Signal	VW			EW		
	Annual SR	Avg Return	CAPM α	Annual SR	Avg Return	CAPM α
Nonlinearity-adjusted Forecast	1.0888	0.0201*** (3.01)	0.0189*** (2.78)	0.9883	0.0179*** (2.74)	0.0173** (2.53)
Base Target	-0.0663	-0.0010 (-0.18)	-0.0030 (-0.61)	-0.0198	-0.0003 (-0.05)	-0.0028 (-0.63)
Predicted Forecast Error	0.9645	0.0167*** (2.67)	0.0157** (2.27)	0.8895	0.0149** (2.46)	0.0145** (2.16)

Thus, analyst base-case forecasts, taken at face value, do not exhibit meaningful cross-sectional return predictability.

By contrast, the Nonlinearity-adjusted Forecast delivers economically large and statistically significant abnormal returns. The VW long-short portfolio earns an average monthly return of 2.01% ($t = 3.01$) and a CAPM alpha of 1.89% ($t = 2.78$), with an annualized Sharpe ratio of 1.09. The EW portfolio earns 1.79% per month ($t = 2.74$) with a CAPM alpha of 1.73% ($t = 2.53$) and an annualized Sharpe ratio of 0.99. The Predicted Forecast Error signal, which sorts on the RFF-predicted correction alone, also generates significant positive returns, but its performance is modestly weaker than that of the combined nonlinearity-adjusted forecast. Its CAPM alpha is 1.57% for VW portfolios and 1.45% for EW portfolios. These results indicate that the nonlinear bias correction contains substantial return-relevant information. At the same time, the stronger performance of the adjusted forecast suggests that the analyst's base-case forecast still contributes useful information once its predictable bias is corrected.

Figure 9 traces how portfolio performance varies with model complexity c , defined as the

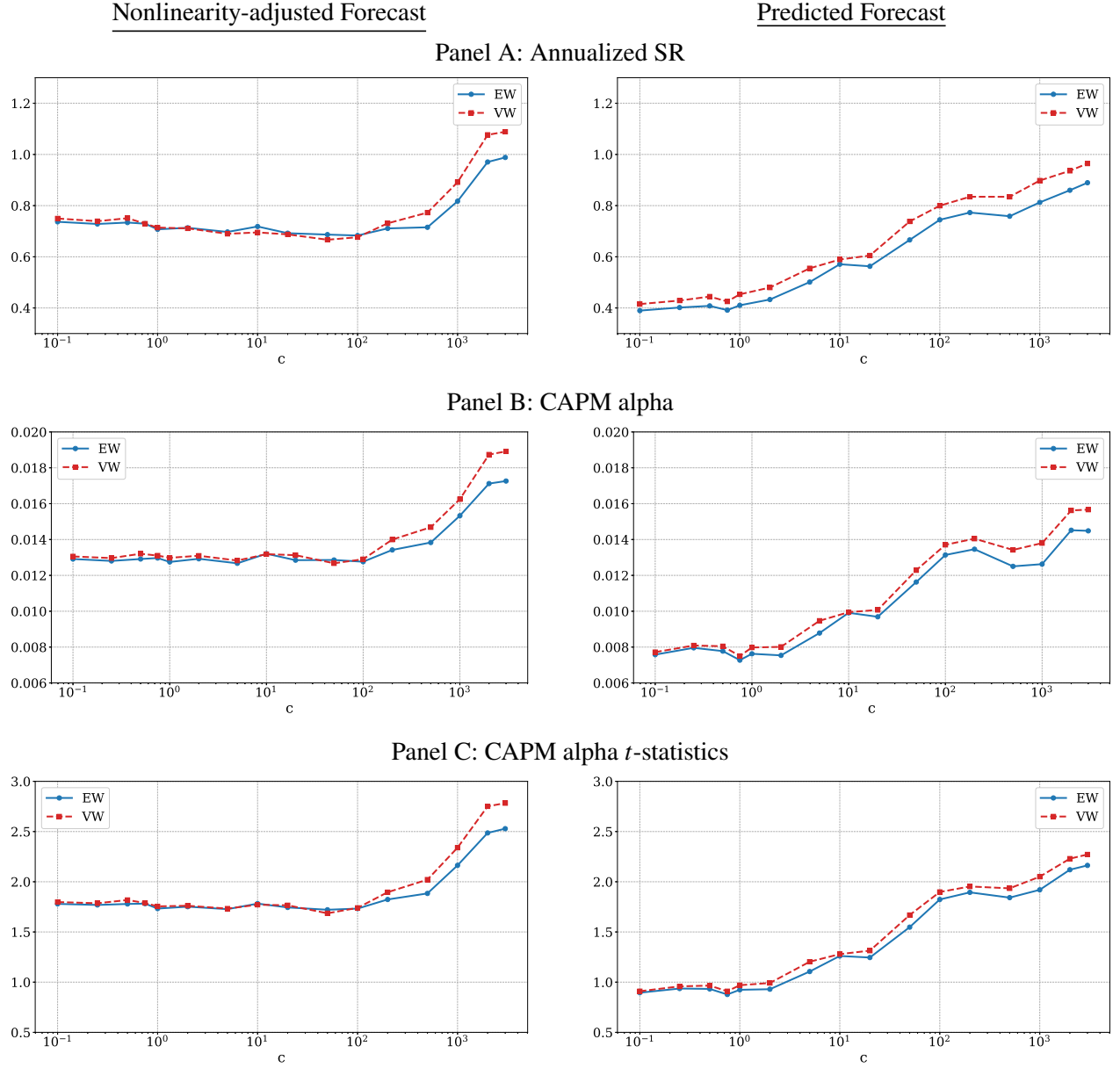


Figure 9 Out-of-sample Long-short Portfolio Performance.

Note: This figure shows the out-of-sample long-short portfolio performance when the model complexity c increases. The training window is 60 observations for each firm, and RFF count P ranges from 6 to 180,000 with bandwidth $\gamma = 2$ and shrinkage $z = 1000$.

number of RFF features divided by the 60-observation training window. For the Nonlinearity-adjusted Forecast, shown in the left column, both the annualized Sharpe ratio and CAPM alpha increase with model complexity, reaching a VW Sharpe ratio of approximately 1.09 and CAPM t -statistics above 2.5 near $c = 3,000$. The Predicted Forecast Error signal, shown in the right column,

exhibits a similar monotone pattern, although at modestly lower levels. Importantly, CAPM alpha is close to zero at low levels of complexity, where the RFF feature space is too sparse to capture the relevant state-dependent interactions. This pattern mirrors the weak explanatory power of the linear specification in Table 4. As model complexity increases, the RFF approximation becomes richer and recovers more of the nonlinear predictive content embedded in analyst narratives.

Across these specifications, the results show that Macro Bear Bias has economically meaningful return implications, but that these implications are difficult to detect using a linear specification. A linear model compresses state-dependent forecast distortions into an average slope that explains little variation. The RFF specification instead recovers nonlinear interactions between macro-heavy bear narratives and market conditions, producing substantial out-of-sample portfolio performance.¹¹

4 Mechanisms

We test two candidate mechanisms for Macro Bear Bias. The first is cognitive: the availability heuristic predicts that analysts anchor bear-case narratives to salient macro-crisis templates because such templates are vivid, widely shared, and easy to retrieve. The second is strategic: career concerns predict that analysts emphasize macro risk in bear scenarios because doing so is professionally safer than making falsifiable firm-specific claims. The evidence supports availability; career concerns receive no support.

4.1 Availability Heuristic

The availability heuristic (Tversky and Kahneman, 1973, 1974) holds that people assess the probability of an event by the ease with which instances come to mind: events that are vivid, widely shared, or recently experienced are recalled more readily, inflating their judged likelihood.

¹¹We report return-based evidence (alphas, Sharpe ratios) rather than direct forecast-accuracy measures such as out-of-sample MSE. The portfolio results show that the predicted bias correction is return-relevant; we do not claim that the RFF model lowers raw forecast-error loss, which is a stricter object than cross-sectional ranking accuracy.

4.1.1 Identification

If Macro Bear Bias arises through this channel, then macro emphasis in bear narratives should respond systematically to factors that heighten the cognitive accessibility of macro-crisis templates. We test this via four predictions: one at the topic level using news coverage shocks (Table 6), and three cross-sectional predictions at the analyst and industry level (Table 7).

Topic salience. The availability heuristic predicts that a sudden burst of news coverage of a macro topic should make that topic cognitively accessible, causing analysts to draw on it more readily when constructing bear-case narratives.

We test this prediction at the topic-month level by regressing bear-case analyst attention to topic k at time t on the WSJ topic shocks, its interaction with WSJ sentiment, and control WSJ sentiment and lagged WSJ attentions:

$$\begin{aligned} \text{BearAttention}_{t,k} = & \beta_1 \hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}} + \beta_2 \hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}} \times \text{WSJSentiment}_{t-\tau,k} \\ & + \beta_3 \text{WSJSentiment}_{t-\tau,k} + \sum_{j=1}^3 \phi_j \text{WSJAttention}_{t-j,k} + \alpha_t + u_{t,k}, \end{aligned} \quad (14)$$

where $\hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}}$ denotes the autoregressive (AR) residual of WSJ attention to topic k at lag τ , capturing the unexpected component of topic attention.¹² $\text{WSJSentiment}_{t-\tau,k}$ is the average WSJ news sentiment for topic k at lag τ , and α_t is the time fixed effect. The three lags of $\text{WSJAttention}_{t-j,k}$ control for persistence in coverage so that $\hat{\varepsilon}_{t-\tau,k}^{\text{WSJAttention}}$ isolates only the transient, unexpected increase in attention. The specification nests two predictions: a level effect ($\beta_1 > 0$), whereby coverage surprises raise analyst bear-case attention, and an asymmetry effect ($\beta_2 < 0$), whereby negative-sentiment coverage amplifies this response.

Table 6 reports results for $\tau \in \{1, 2, 3, 4\}$. The level effect is positive and statistically significant at short lags: a one-unit attention shock raises bear-case macro attention by 0.064 percentage points at $\tau = 1$ ($t = 1.85$) and 0.052 points at $\tau = 2$ ($t = 1.72$), declining to an insignificant 0.042 and

¹²We use an AR(3) model for each topic, since the Akaike information criterion (AIC) selects AR(3) for 40 of the 52 topics.

Table 6 Bear-Case Macro Attention and Topic Salience

Note: This table reports panel regressions of bear-case analyst attention to topic k at time t on the autoregressive (AR) residual of WSJ news coverage (attention shock), its interaction with WSJ news sentiment, the sentiment level, and three lags of WSJ topic attention; see Equation (14). Columns correspond to lag $\tau \in \{1, 2, 3, 4\}$. All specifications include time fixed effects. t -statistics in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	$\tau = 1$	$\tau = 2$	$\tau = 3$	$\tau = 4$
WSJ Attention Shock $_{t-\tau,k}$	0.0642* (1.85)	0.0524* (1.72)	0.0417 (1.54)	0.0207 (1.50)
WSJ Attention Shock $_{t-\tau,k} \times$ WSJ Sentiment $_{t-\tau,k}$	-0.1876* (-1.81)	-0.2374** (-2.43)	-0.1855** (-2.10)	-0.0377 (-0.55)
WSJ Sentiment $_{t-\tau,k}$	0.0001 (0.72)	0.0000 (0.24)	0.0001 (0.56)	0.0001 (0.64)
WSJ Attention $_{t-1,k}$	0.3085*** (11.02)	0.2998*** (10.87)	0.3004*** (10.48)	0.3011*** (10.14)
WSJ Attention $_{t-2,k}$	0.2835*** (10.33)	0.2868*** (10.44)	0.2862*** (10.20)	0.2867*** (10.16)
WSJ Attention $_{t-3,k}$	0.3826*** (11.94)	0.3876*** (12.58)	0.3873*** (12.12)	0.3865*** (11.68)
Time FE	Yes	Yes	Yes	Yes
Observations	11,440	11,388	11,336	11,284
R^2	88.95%	89.00%	89.00%	88.98%

0.021 at $\tau = 3$ and $\tau = 4$. The interaction coefficient is negative and statistically significant through $\tau = 3$: $\hat{\beta}_2 = -0.188$, -0.237 , and -0.186 at $\tau = 1, 2$, and 3 (t -statistics -1.81 , -2.43 , -2.10), attenuating to an insignificant -0.038 at $\tau = 4$. Baseline sentiment alone is economically negligible and statistically insignificant across all columns, indicating that the direction of coverage matters only in amplifying surprises, not independently.

These results are consistent with the availability heuristic. Unexpected increases in macro news coverage shift bear-case analyst attention toward the newly salient topic, and this shift is amplified when coverage is negative—precisely the threatening, crisis-type signal identified by the availability literature as most cognitively accessible (Tversky and Kahneman, 1973). The attenuation of both effects at $\tau \geq 3$ mirrors the transient nature of news salience: availability fades as coverage recedes.

Industry-crisis salience. A direct cross-sectional prediction of the availability heuristic is that crisis templates should be deployed differentially across industries, depending on the perceived

relevance of a given macro crisis to each sector. The 2008 Global Financial Crisis was widely framed as a systematic-risk event centered on financial contagion, so the availability of macro-downside narratives should have increased most for firms in high-systematic-exposure sectors (e.g., financials, energy, and materials) relative to defensive sectors such as utilities, consumer staples, and healthcare. We test this by interacting a post-GFC indicator with a high-systematic-industry dummy.

Column (1) of Table 7 reports the results. The interaction is economically and statistically significant: after the GFC, MacroGap increases by 4.9 percentage points more for firms in high-systematic sectors than for other firms ($t = 2.61$).¹³ Combining the main effect and the interaction, high-systematic sectors had on average -8.9 percentage-point *lower* MacroGap than other sectors before the GFC; post-GFC, the differential narrows by 4.9 points to about -4.0 percentage points before any common post-GFC effect is added. The crisis therefore did not flip high-systematic sectors to higher MacroGap; it narrowed a pre-existing negative gap, which is the within-firm pattern the availability heuristic predicts when the crisis differentially raises the cognitive accessibility of macro-downside templates for these sectors.¹⁴

Analyst crisis experience. The next three tests require an analyst-level identifier and therefore restrict the sample to the *analyst-matched subsample*. We match each Morgan Stanley report to IBES via the analyst’s last name (cleaned of titles and suffixes) and the covered firm’s PERMNO within a 90-day window around the report date; a report enters the matched subsample if at least one IBES analyst record satisfies the join, and we retain the longest-tenured matching analyst when

¹³The High Systematic main effect of -8.9 ($t = -4.30$) reported in column (1) is identified from the entry-and-exit composition of the unbalanced panel rather than from within-firm variation, since GICS sector classification is essentially time-invariant within firm; we verify this in unreported tests by repeating column (1) with industry fixed effects in place of firm fixed effects and obtaining the same interaction coefficient. The Post-GFC $\times$ High Systematic interaction is properly identified under firm fixed effects regardless.

¹⁴We also test whether the post-COVID increase in MacroGap is concentrated in COVID-sensitive industries (airlines, hotels, retail). It is not: the COVID interaction with a COVID-sensitive industry indicator is essentially zero (-0.06 , $t = -0.06$). This contrast suggests that COVID activated a universal “shutdown” narrative across all sectors, whereas the GFC activated a sector-specific “financial contagion” template.

multiple candidates appear. The matched subsample contains 17,257 report observations, a 45% reduction from the full panel of 31,445 observations used in column (1).¹⁵

If availability operates through personal exposure to crisis episodes, then analysts active during the GFC should carry stronger macro-crisis anchors. We construct two GFC-experience indicators: one for analysts with a record at Morgan Stanley before March 2009, and one for analysts with a record at *any* brokerage before that date, each proxied by the analyst's earliest report date in the respective database. This pair of indicators distinguishes two channels: the MS-specific measure captures exposure to the firm's crisis-era narrative conventions, while the profession-wide measure captures generic crisis memory.

Columns (2) and (3) of Table 7 report the results. Analysts who experienced the GFC at Morgan Stanley exhibit a 1.2-point higher MacroGap ($t = 1.65$); analysts who experienced the GFC elsewhere show no discernible effect. The divergence is consistent with availability operating through firm-specific narrative templates: it is not crisis memory per se that elevates macro emphasis, but exposure to a particular institution's crisis-era framing conventions.¹⁶

Sector-level narrative herding. A final prediction follows from the shared information environment of sector analysts. If analysts covering firms in the same sector draw on the same pool of salient crisis templates—through common media exposure, industry conferences, and peer discussion—their MacroGap values should covary within sector-quarters, even after absorbing firm fixed effects. We construct a leave-one-out (LOO) sector-quarter average of MacroGap and test whether an analyst's own MacroGap covaries with this peer average.

¹⁵The reduction is driven primarily by IBES coverage gaps (about 32%) and ambiguous name matches that the join rule discards conservatively (about 13%). Matched and unmatched reports differ on observable dimensions of analyst tenure and firm coverage: matched reports skew toward larger-cap, longer-tenured analyst coverage, which is exactly the segment in which the GFC-experience and IB-relationship tests have the most variation. We flag this as a caveat for the career-concerns null in Section 4.2: power against career-concern effects operating predominantly among junior or less-visible analysts is limited by the matching procedure. A full balance table by matching status is in preparation; the current draft documents the procedure here and reports the cross-tab of MacroGap, analyst tenure, and IB-relationship frequency in matched versus unmatched observations in a future revision.

¹⁶We also examine whether the MS-specific GFC effect reflects generic MS tenure rather than crisis-specific experience. In a horse-race regression including both the GFC-experience indicator and years of MS tenure, the GFC coefficient remains positive (+1.27) while the tenure coefficient is essentially zero (-0.02 , $t = -0.14$). The point estimate is directionally consistent but imprecisely estimated in the horse-race specification; we therefore read the standalone result as suggestive rather than definitive.

Column (4) of Table 7 is consistent with this prediction. The leave-one-out sector MacroGap positively predicts the analyst’s own MacroGap (0.083, $t = 1.75$), suggesting that analysts covering firms in the same sector–quarter converge on similar levels of macro emphasis beyond what firm fixed effects explain.

Considered jointly, the four tests in Tables 6 and 7 are consistent with an availability-heuristic channel behind Macro Bear Bias. Bear-case macro attention rises in response to negative news coverage surprises, is larger in high-systematic sectors following the GFC, is elevated for analysts who experienced the crisis at Morgan Stanley, and co-moves within sector–quarter peer groups—a pattern consistent with shared narrative templates rather than independent judgment. We note that three of the four point estimates are significant at conventional one-sided 5% levels (the WSJ attention shock at $\tau = 1$, the MS-specific GFC indicator, and the leave-one-out sector MacroGap fall between $t = 1.65$ and $t = 1.85$), with the post-GFC $\times$ high-systematic interaction clearing two-sided 5%; we adopt one-sided tests because the availability mechanism delivers signed directional predictions ($\beta_1 > 0$, $\beta_2 < 0$, positive sector covariance), and disclose the convention here.

4.1.2 Directional Boldness and Narrative Coherence

The availability heuristic explains why analysts default to macro-heavy bear narratives. It does not identify which analysts are most susceptible. A simple narrative-coherence prediction follows from the heuristic itself: the structure of the bear case should depend on what is *not* already supplied by the central scenario.

Simply put: an analyst whose base-case price target exceeds the consensus has committed to a bullish firm-specific thesis. When the Morgan Stanley Risk-Reward framework requires this analyst to construct a bear case, the most cognitively available source of downside is external—macroeconomic deterioration, recession, or broad multiple compression—because the firm-specific view is favorable. An analyst whose base-case target falls below the consensus has already embedded firm-specific problems into the central scenario (execution risk, competitive threats, margin pressure); idiosyncratic downside is already supplied, and macro narratives are less salient.

We test this prediction by defining “directional boldness”: an indicator equal to one if the

Table 7 Availability Heuristic: Determinants of Macro Bear Bias

Note: This table reports panel regressions of MacroGap (bear minus base macro attention, 0–100 scale) on proxies for the availability heuristic. Column (1) interacts a post-GFC indicator with a high-systematic-industry dummy (financials, energy, materials). Columns (2)–(3) test whether analysts who experienced the GFC at Morgan Stanley (col 2) or at any brokerage (col 3) exhibit larger MacroGap. Column (4) tests sector-level narrative herding using a leave-one-out sector–quarter average of MacroGap. Columns (2)–(3) are restricted to the analyst-matched subsample. All specifications include firm and year fixed effects. Standard errors are clustered by firm. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroGap			
	(1)	(2)	(3)	(4)
Post-GFC × High Systematic	4.9229*** (2.61)			
High Systematic	−8.9346*** (−4.30)			
GFC Experienced (MS)		1.1544* (1.65)		
GFC Experienced (Prof.)			0.1594 (0.25)	
LOO Sector MacroGap				0.0828* (1.75)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	31,445	17,257	17,257	30,979
R^2	0.0008	0.0007	0.0000	0.0010

analyst’s base-case price target exceeds the IBES consensus price target (“bold above”), and a second indicator equal to one if the base-case target falls below the consensus (“bold below”). We match each Morgan Stanley report to the most recent IBES consensus price target for the same firm within 90 days, yielding a matched sample of 26,858 observations.

Table 8 reports the results. Columns (1)–(2) report the two indicators separately: analysts whose base case exceeds consensus exhibit a MacroGap 1.05 percentage points above the sample mean ($t = 4.47$); analysts whose base case falls below consensus exhibit a MacroGap 1.09 percentage points below the mean ($t = -4.66$). Column (3) includes both indicators simultaneously, so that the omitted category becomes the set of analysts whose base-case target equals the IBES consensus. In this specification the asymmetry sharpens: Bold-Below survives (-1.12 , $t = -1.65$)

while Bold-Above collapses to economic zero (-0.03 , $t = -0.04$). The cross-section is therefore driven primarily by below-consensus analysts writing *less* macro-heavy bear narratives—not by above-consensus analysts writing more macro-heavy ones. Columns (4)–(5) replace the indicators with continuous boldness measures—the signed deviation $(P_{MS}^{\text{base}} - P^{\text{cons}})/P^{\text{cons}}$ and its absolute value. Neither continuous measure is statistically significant, and the within- R^2 falls to essentially zero: the relationship is not a smooth slope through consensus but an asymmetric kink at the below-consensus margin.

This pattern is consistent with the availability heuristic operating in scenario construction. When the firm-specific thesis is already bearish, the firm-specific channel supplies the downside and the bear case is less macro-heavy; when the firm-specific thesis is bullish or at consensus, macro narratives carry the bear case but are not differentially elevated relative to the at-consensus reference group. The cross-section is therefore a fingerprint of how analysts allocate downside material across scenarios. It does not directly identify the discount-rate contamination channel of Section 3—which is a price-level prediction about how macro-heavy bear narratives inflate $\tilde{\beta}_i$ and depress base-case targets—because that channel requires a timing design (e.g., MacroGap lagged into the next report’s base-case deviation) that is beyond the scope of this cross-section. The forecast-error evidence in Section 2.4 provides the price-level test; the cross-section here identifies the narrative-coherence margin on which the bias varies across analysts.

One might object that this pattern is mechanical: a bullish analyst has “used up” the firm-specific upside and must turn to macro factors for the downside. But firm-specific downside is always available—execution risk, competitive entry, regulatory changes, management turnover, product delays—regardless of the base-case outlook. The systematic tendency to emphasize macro over these alternatives, conditional on base-case direction, is itself the bias. Moreover, if the pattern were purely mechanical, bearish analysts should exhibit the mirror image: they should need macro factors in their *bull* case. But as documented in Section 2.4, the macro attention gap in bull narratives does not predict forecast errors. The asymmetry is bear-case-specific, consistent with a psychological mechanism rather than a mechanical artifact.

Table 8 Directional Boldness and Macro Bear Bias

Note: This table reports panel regressions of MacroGap (bear minus base macro attention, 0–100 scale) on boldness measures. Bold above (below) consensus equals one if the Morgan Stanley base-case price target is above (below) the IBES consensus price target. In column (3), both indicators are included, so the omitted category is the set of observations for which the Morgan Stanley base-case target equals the IBES consensus (subject to the unit-price rounding in IBES). Column (4) replaces the indicators with the signed continuous deviation $(P_{MS}^{base} - P^{cons})/P^{cons}$, column (5) uses its absolute value, a Hong–Kubik-style conviction measure. The sample is restricted to observations with an available IBES consensus match within 90 days. All specifications include firm and year fixed effects. Standard errors are clustered by firm. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroGap				
	(1)	(2)	(3)	(4)	(5)
Bold Above Consensus	1.0491*** (4.47)		−0.0299 (−0.04)		
Bold Below Consensus		−1.0938*** (−4.66)	−1.1221* (−1.65)		
Bold Signed				0.0865 (1.34)	
Bold Absolute					0.0428 (0.61)
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Observations	26,858	26,858	26,858	26,380	26,380
R^2	0.0012	0.0013	0.0013	0.0000	0.0000

4.2 Career Concerns

A natural competing explanation for Macro Bear Bias is strategic rather than cognitive. Sell-side analysts face asymmetric reputational costs: firm-specific errors are falsifiable and attributable, while consensus-correlated errors provide reputational cover (Hong and Kubik, 2003). This asymmetry could generate a “cover your asset” (CYA) heuristic: when constructing downside scenarios, analysts may emphasize macro risks because macro-based bear narratives are difficult to falsify, relationally neutral toward management, and consensus-aligned. Under this view, Macro Bear Bias reflects an incentive-compatible communication strategy rather than a cognitive distortion.

If career incentives drive macro emphasis, three predictions follow that our proxies can test: (i) junior analysts, who face greater reputational risk from differentiated misses, should exhibit

larger MacroGap; (ii) firms with investment-banking relationships with Morgan Stanley should show larger MacroGap, since firm-specific bear narratives are relationally costly; and (iii) analyst identity should explain a substantial share of MacroGap variation. The data provide little support for (i) and (ii) and only partial support for (iii). A fourth implication—that analysts who deviate more from the consensus should show smaller MacroGap—requires a continuous consensus-deviation regressor that we do not estimate in the current draft; we therefore restrict our conclusions to the proxies actually tested.

Appendix Table A.4 reports regressions of MacroGap on analyst experience, measured both as Morgan Stanley tenure and as years of industry experience across all brokerages (using the full IBES price-target detail file to identify each analyst’s career start at any firm). Career concerns predict that junior analysts exhibit larger MacroGap; the data contradict this. The MS-tenure junior indicator enters with a *negative* coefficient (-0.69 , $t = -1.66$): analysts new to Morgan Stanley show *less* macro emphasis in bear narratives, not more. The profession-wide junior indicator is statistically indistinguishable from zero. The negative MS-tenure result is more naturally read through the availability lens, with the caveat that availability in our setting operates through both personal recall of crisis episodes and repeated exposure to firm-specific narrative templates. Analysts new to Morgan Stanley have not yet been steeped in the firm’s macro-heavy bear-case template; as they accumulate tenure, the template becomes cognitively more accessible. This is the institutional-repetition leg of availability, not personal-memory availability in the original Tversky–Kahneman sense. Our tests cannot discriminate between the two within a single-brokerage sample, and we do not claim to: the GFC-at-MS finding in Table 7 is consistent with both personal crisis memory and institution-specific socialization into the firm’s crisis-era framing. We read both legs as instances of the availability channel and discuss the limits of this identification in Section 5.

If career concerns operate through relational incentives, then analysts covering firms with active Morgan Stanley investment-banking relationships should show larger MacroGap, since firm-specific bear narratives risk antagonizing management during the advisory relationship. We construct an IB-relationship indicator equal to one if Morgan Stanley served as bookrunner or lead manager on a new issuance for the covered firm within a specified lookback window, using the LSEG SDC New

Issues database. Table 9, columns (1)–(4), reports the results across four lookback windows (6, 12, 24, and 36 months). All coefficients are positive, consistent with the predicted direction, but none approaches statistical significance (t -statistics range from 0.46 to 1.00).¹⁷

Columns (5)–(7) of Table 9 test three additional proxies motivated by the career-concerns literature. If analysts who cover more firms rely on generic templates, coverage breadth should predict higher MacroGap (column 5: -0.04 , $t = -0.80$). If analysts who write more reports entrench house templates, report frequency should predict higher MacroGap (column 6: -0.01 , $t = -0.52$). If analysts with poor historical forecast accuracy use macro framing as reputational insurance, average forecast error should predict higher MacroGap (column 7: 0.48 , $t = 0.78$). None of these proxies is statistically significant, and the coverage and frequency measures carry signs opposite to the career-concerns prediction.¹⁸

On balance, none of the career-concern proxies—analyst experience, IB relationships, coverage breadth, report frequency, forecast accuracy—predicts MacroGap in the direction the channel implies. The only marginally significant coefficient, junior MS tenure, carries the wrong sign. We read this as little support for the career-concerns channel through the proxies available to us; we do not claim the channel is ruled out, since institutional-client demand, sector-head review, internal compliance, and management-access pressure are not captured by these measures and could in principle generate Macro Bear Bias through routes our data do not observe.¹⁹ Availability, by contrast, is supported across the industry–crisis interaction, GFC-experience, sector-herding, and directional-boldness tests.

¹⁷The SDC data match 34.5% of deals to our panel via CUSIP-to-PERMNO mapping, yielding IB-relationship rates of 12% (6-month window) to 39% (36-month window). The null result is unlikely to be driven by insufficient variation in the IB indicator.

¹⁸We also examine whether analyst identity explains a meaningful share of MacroGap variation by adding 232 analyst fixed effects to a specification with firm and year fixed effects (Appendix Table A.5). The within- R^2 increases by 2.05 percentage points (from 0.28% to 2.33%), indicating persistent heterogeneity in macro-narrative style across analysts. This is consistent with either career incentives or cognitive differences in scenario construction, and does not uniquely identify career concerns.

¹⁹The proxies in Table 9 have within- R^2 values close to zero, so the tests have limited power against small but economically meaningful career-concern effects. A formal power analysis would help bound the magnitude of effect that the IB-relationship coefficient is consistent with; we leave this calculation to a future revision.

Table 9 Career Concern Proxies: IB Relationships and Analyst Characteristics

Note: This table reports panel regressions of MacroGap (bear minus base macro attention, 0–100 scale) on career-concern proxies. Columns (1)–(4) test whether Morgan Stanley investment-banking relationships predict MacroGap, using lookback windows of 6, 12, 24, and 36 months. IB relationship equals one if MS served as bookrunner or lead manager on a new issuance for the covered firm within the indicated window (source: SDC New Issues). Coverage equals the number of firms the analyst covers in the same calendar year. Reports equals the analyst’s annual report count. Accuracy equals the analyst’s mean absolute forecast error. Columns (5)–(7) are restricted to the analyst-matched subsample. All specifications include firm and year fixed effects. Standard errors are clustered by firm. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroGap						
	IB 6m	IB 12m	IB 24m	IB 36m	Coverage	Reports	Accuracy
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
MS IB Relationship	0.1484 (0.46)	0.3363 (1.00)	0.2481 (0.59)	0.3474 (0.70)			
Firms Covered					−0.0351 (−0.80)		
Reports / Year						−0.0055 (−0.52)	
Avg. Forecast Error							0.4776 (0.78)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31,445	31,445	31,445	31,445	17,257	17,257	17,255
R^2	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000

5 Conclusion

We document a systematic pattern in how analysts narrate downside risk. Using scenario-based analyst reports, we show that narratives contain information that is not visible from numerical forecasts alone. Analysts do not merely forecast prices; they also explain why different outcomes may occur. These explanations reveal a systematic distortion in subjective risk attribution: bear-case scenarios are framed as disproportionately macro-driven, even when realized outcomes do not support a larger macro component of downside risk.

We call this pattern Macro Bear Bias. The key insight is not that macro risk is unimportant, nor that analysts are wrong to discuss macro conditions in downside states. Rather, the bias lies in the failure to condition correctly. Conditional on the same subsequent market environment,

realized CAPM R^2 is similar across realized bear, base, and bull outcomes, yet bear narratives remain persistently more macro-heavy than base or bull narratives. Downside risk is therefore not only perceived as bad; it is narrated as systematically macro in nature.

This narrative distortion matters economically. The bear–base macro-attention gap predicts subsequent errors in analysts’ base-case forecasts, suggesting that macro-heavy downside narratives spill over into central valuation judgments and make analysts too pessimistic *ex ante*. The predictive relation is especially strong in nonlinear specifications, consistent with the idea that the cost of a fixed macro-downside template depends on the market environment in which it is applied.

Mechanism tests are consistent with an availability-heuristic channel for Macro Bear Bias. As predicted by this mechanism, macro-heavy bear narratives respond to negative macro-news shocks, concentrate in crisis-exposed settings, and converge among analysts facing shared sector-level narrative environments. The economic realization is consistent with a discount-rate contamination channel—salient macro-crisis templates inflate perceived systematic risk in bear scenarios, raising the discount rate used in valuation and depressing base-case targets—though our forecast-error evidence does not decompose the bias into discount-rate and cash-flow components, and a cash-flow-revision channel cannot be ruled out without that decomposition. By contrast, the simple career-concerns proxies we observe carry little explanatory power; we read this as no support for that channel through these proxies rather than as a wholesale rejection of strategic explanations.

The broader impact of this paper is to show that narratives are not merely decorative explanations attached to forecasts; they are an economically important part of belief formation in financial markets. By revealing how sophisticated intermediaries systematically misattribute downside risk to macro forces, our findings highlight a new channel through which narrative templates can shape valuations, forecast errors, and ultimately market beliefs. The paper therefore contributes to a broader understanding of price formation: distortions need not arise only from biased numerical expectations, but can also emerge from the way agents organize and communicate uncertainty. Methodologically, the paper shows how combining structured scenario reports, LLM-based narrative measurement, and realized return benchmarks can open a new empirical window into subjective risk perception.

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INTERNET APPENDIX

A LLM Prompt

A.1 Subjective Risks Identification Prompt

You are a financial analyst trained in equity valuation and empirical asset pricing. You will analyze the Bull, Base, and Bear case texts from a Morgan Stanley equity research report. Your task is to extract reasoning, classify it as firm-level, industry-level, or macro-level, identify the economic channel, and quantify attention shares across cases. Write in a concise, research-style tone used in equity research reports.

Return your full answer ****strictly in JSON format****, with one JSON object per report (suitable for .jsonl storage). Each JSON object must include keys: "filename", "bull_case", "base_case", "bear_case", and "cross_case_summary". Do not include explanations outside the JSON.

Tasks:

- 1.Price Targets: List the bull case, base case, and bear case price targets stated in the Risk Reward section.
- 2.Reason Extraction: Identify all distinct reasons for the bull, base, and bear cases. Requirements:
 - 2.1 Include any explicitly stated upside risks as part of the bull-case reasons.
 - 2.2 Include any explicitly stated downside risks as part of the bear-case reasons.
 - 2.3 Preserve the original wording as closely as possible (avoid paraphrasing).
 - 2.4 Number each reason sequentially (e.g., 1., 2., 3.).
 - 2.5 Label each reason as one of the following categories: Firm, Industry, Macro, or Mixed. If labeled Mixed, specify which component (Firm, Industry, or Macro) is dominant.
 - 2.6 Assign each reason to a single economic channel that best represents its

underlying mechanism, such as expected cash flow, discount rate, risk aversion or asset demand.

- 3.Attention Quantification: Quantify the analyst's relative attention across Firm, Industry, and Macro news as: (Firm %, Industry %, Macro %), ensuring $\text{Firm \%} + \text{Industry \%} + \text{Macro \%} \leq 100$.
- 4.Attention Evolution Summary: Summarize how the analyst's attention distribution across Firm, Industry, and Macro news (from Step 2.5) evolves across the bull, base, and bear cases.
- 5.Channel Evolution Summary: Summarize how the dominant economic channels (from Step 2.6) evolve across the bull, base, and bear cases.

Output format (strict JSON):

```
{
  "filename": "<report_name>",
  "bull_case": {
    "bull_price_target": "<numerical price target or range>",
    "reasons": [
      {
        "id": 1,
        "text": "<verbatim reason>",
        "category": "Firm | Industry | Macro | Mixed (dominant: Firm/Industry/Macro)",
        "economic_channel": "Expected Cash Flow | Discount Rate | Risk Aversion | Asset Demand"
      }
    ],
    "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
    "analytical_summary": "[one concise sentence summarizing key focus of bull case]"
  },
  "base_case": {
    "base_price_target": "<numerical price target or range>",
```

```

"reasons": [
  {
    "id": 1,
    "text": "<verbatim reason>",
    "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
    "economic_channel": "Expected Cash Flow | Discount Rate | Risk
      Aversion | Asset Demand"
  }
],
"attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
"analytical_summary": "[one concise sentence summarizing key focus of base
  case]"
},

"bear_case": {
  "bear_price_target": "<numerical price target or range>",
  "reasons": [
    {
      "id": 1,
      "text": "<verbatim reason>",
      "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
      "economic_channel": "Expected Cash Flow | Discount Rate | Risk
        Aversion | Asset Demand"
    }
  ],
  "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
  "analytical_summary": "[one concise sentence summarizing key focus of bear
    case]"
},

"cross_case_summary": {
  "attention_evolution": "[summary of how Firm/Industry/Macro attention
    shifts across bull -> base -> bear]",

```

```

    "channel_evolution": "[summary of how dominant economic channels shift
        across bull -> base -> bear]"
  }
}

```

Rules:

- All percentages must be integers (no decimals).
- Maintain an analytical and academic tone throughout.
- Do NOT include any text outside the JSON structure.
- Ensure structural consistency and valid JSON formatting.

A.2 Topic Classification Prompt

You are a financial research assistant.

The input summaries already contain ONLY macro-related content.

Do NOT extract or filter anything.

Tasks:

1. Classify each scenario (Bull, Base, Bear) using the taxonomy.
 - Multiple topics allowed.
 - If broad -> "Mixed"
 - If not covered -> "Other: <specific macro theme>"
 - If empty -> "No explicit macro content"
 - Also assign a percentage weight to each topic based on its importance in the summary:
 - The percentages must sum to 100 for each scenario
 - Use integers (e.g., 20, 30, 50)
 - The order of percentages must match the order of topics exactly
1. Compare macro differences:

Bear vs Base:

- If both Bear and Base are empty -> "No explicit macro content"
- If Base is empty but Bear is not:
 - > return ALL Bear topics
- If Bear is empty but Base is not:
 - > "No meaningful difference"
- Otherwise:
 - Identify ONLY macro topics that appear in Bear but NOT in Base
 - Do NOT include topics already in Base
 - If no additional topics -> "No meaningful difference"
- For any returned topics:
 - Assign percentage weights that sum to 100
 - Reflect relative importance of differences
 - Order must match topics exactly

Bull vs Base:

- If both Bull and Base are empty -> "No explicit macro content"
- If Base is empty but Bull is not:
 - > return ALL Bull topics
- If Bull is empty but Base is not:
 - > "No meaningful difference"
- Otherwise:
 - Identify ONLY macro topics that appear in Bull but NOT in Base

- Do NOT include topics already in Base
- If no additional topics -> "No meaningful difference"

- For any returned topics:
 - Assign percentage weights that sum to 100
 - Reflect relative importance of differences
 - Order must match topics exactly

Rules:

- Use only provided text
- No inference beyond text
- Topics must come from taxonomy (+ Mixed / Other)
- "Other" must specify a real macro concept
- Output valid JSON only

Taxonomy:

{TOPIC_TAXONOMY}

Input:

[BULL]

{bull_summary}

[BASE]

{base_summary}

[BEAR]

{bear_summary}

Output JSON:

{{

```

"bull": {{
  "topics": ["..."],
  "percentages": []
}},
"base": {{
  "topics": ["..."],
  "percentages": []
}},
"bear": {{
  "topics": ["..."],
  "percentages": []
}},
"bear_vs_base": {{
  "topics": ["..."],
  "percentages": []
}},
"bull_vs_base": {{
  "topics": ["..."],
  "percentages": []
}}
}}"".strip()

```

A.3 Systematic vs. Idiosyncratic Prompt

You are a financial research analyst.

Your task is to classify each statement based only on its text content.

Determine whether the primary driver described in the statement is:

- Systematic
- Idiosyncratic
- Mixed

Judge only from the text itself.

DEFINITIONS

Systematic:

Primarily driven by broad common factors,
affecting multiple firms or the overall market,
and generally not diversifiable.

Idiosyncratic:

Primarily driven by firm-specific factors,
mainly affecting one firm or a small number of firms,
and generally diversifiable.

Mixed:

Use only when both systematic and idiosyncratic factors
are explicitly stated and equally important.

CRITICAL RULES

1. Primary Driver Rule

Classify based on the main economic driver.

2. Mixed Rule

Use Mixed only when both types are explicitly present
and neither is clearly dominant.

3. Forced Choice Rule

Every statement must be classified as:

- Systematic
- Idiosyncratic
- Mixed

4. Background vs Driver Rule

If broad market or environmental conditions are only background context,
but the company itself drives the outcome,
classify as Idiosyncratic.

INSTRUCTIONS

For each item:

1. Read the statement carefully.
2. Determine the true primary driver.
3. Return one final label.

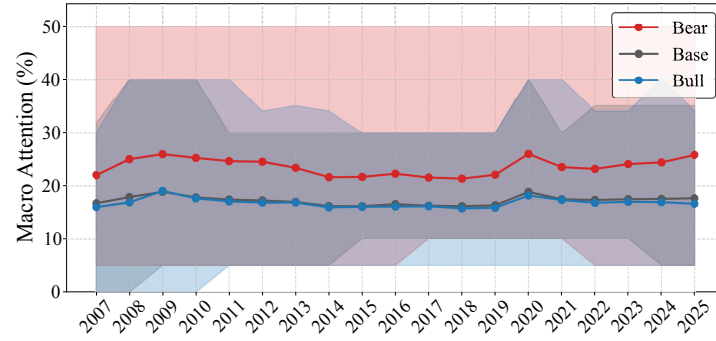
OUTPUT FORMAT

Return ONLY valid JSON:

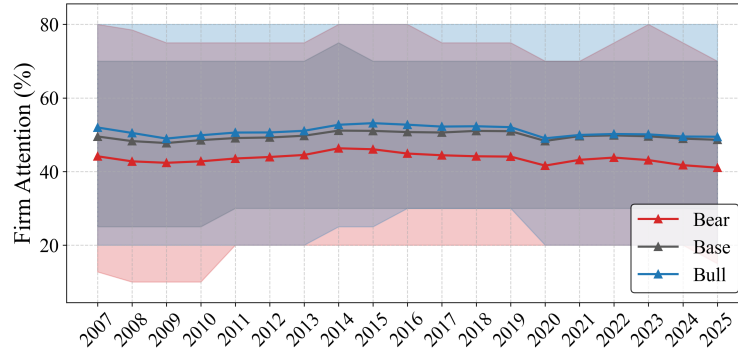
```
{
  "items": [
    {
      "scenario": "bull/base/bear",
      "segment_number": "1",
      "original_text": "...",
      "correct_label": "Systematic",
      "explanation": "short explanation"
    }
  ]
}
```

B Additional Exhibits

Panel A: Macro Attention



Panel B: Firm Attention



Panel C: Industry Attention

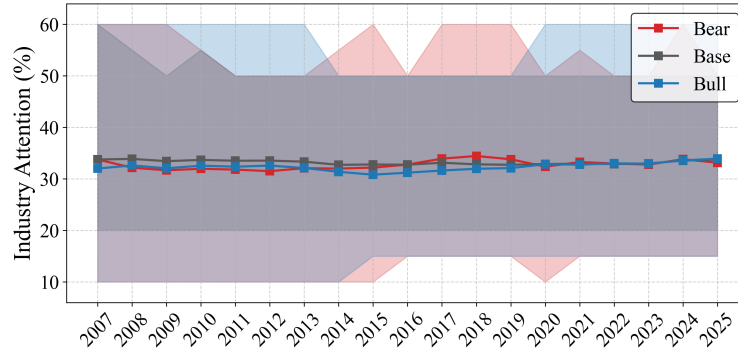


Figure A.1 Time Series of Attention to Firm, Industry, and Macro News.

Note: Time series of annual average attention shares to firm-, industry-, and macro-level news. The solid lines represent yearly means. The shaded areas correspond to the empirical 5th to 95th percentile range of the cross-sectional distribution in each year.

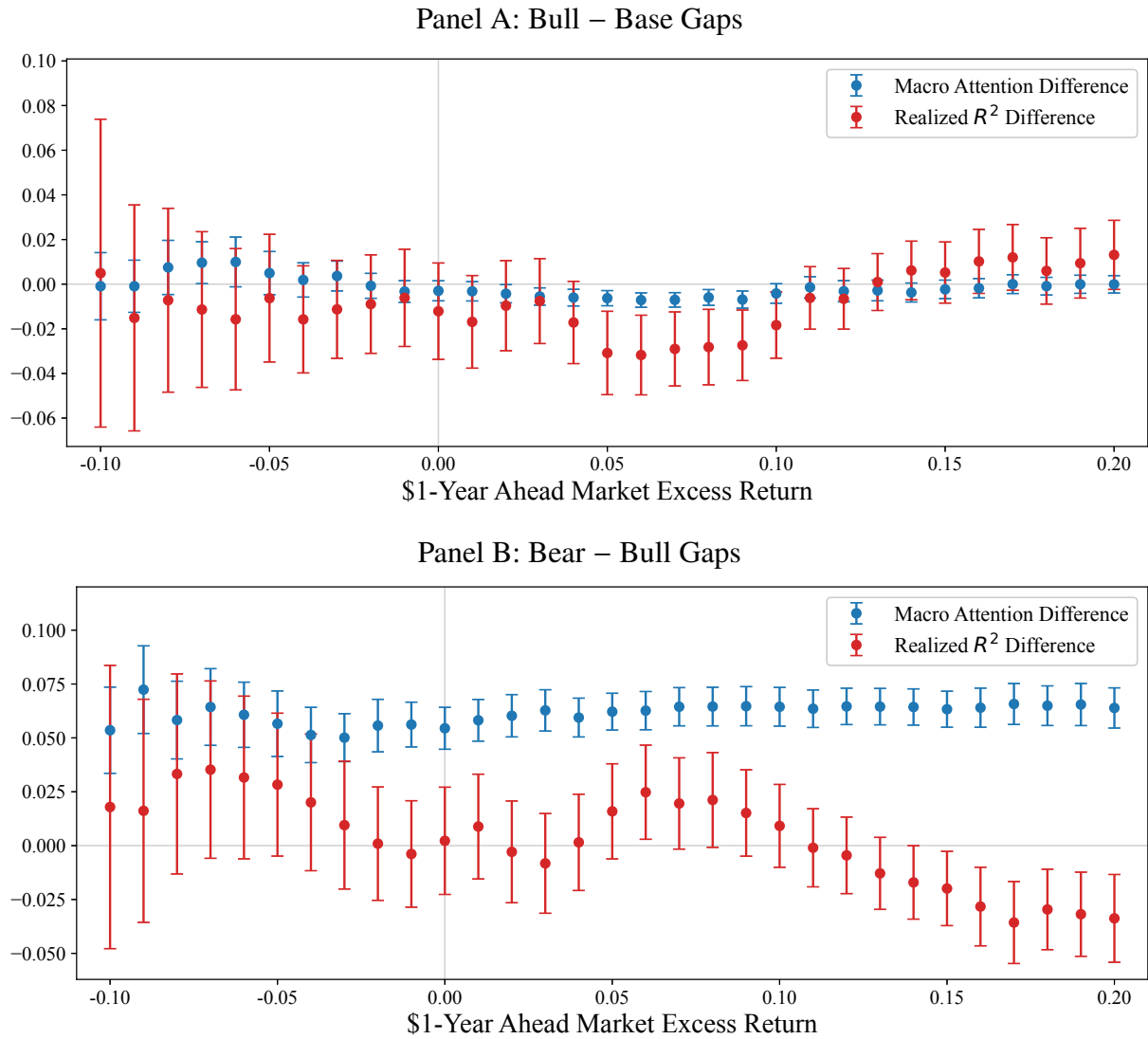


Figure A.2 Gaps in Macro Attention and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bull–base and bear–bull differences in macro attention and realized CAPM R^2 across subsamples defined by 10% bins of forward 252-day market excess returns following the analyst report release. Each point includes reports with $x - 5\% \leq (R_m - R_f) \leq x + 5\%$ and applies the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention in the bull, base, and bear cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bull, realized base, and realized bear cases. The plotted values represent the bull-minus-base and bear-minus-bull differences. Error bars show 95% confidence intervals from constant-only regressions.

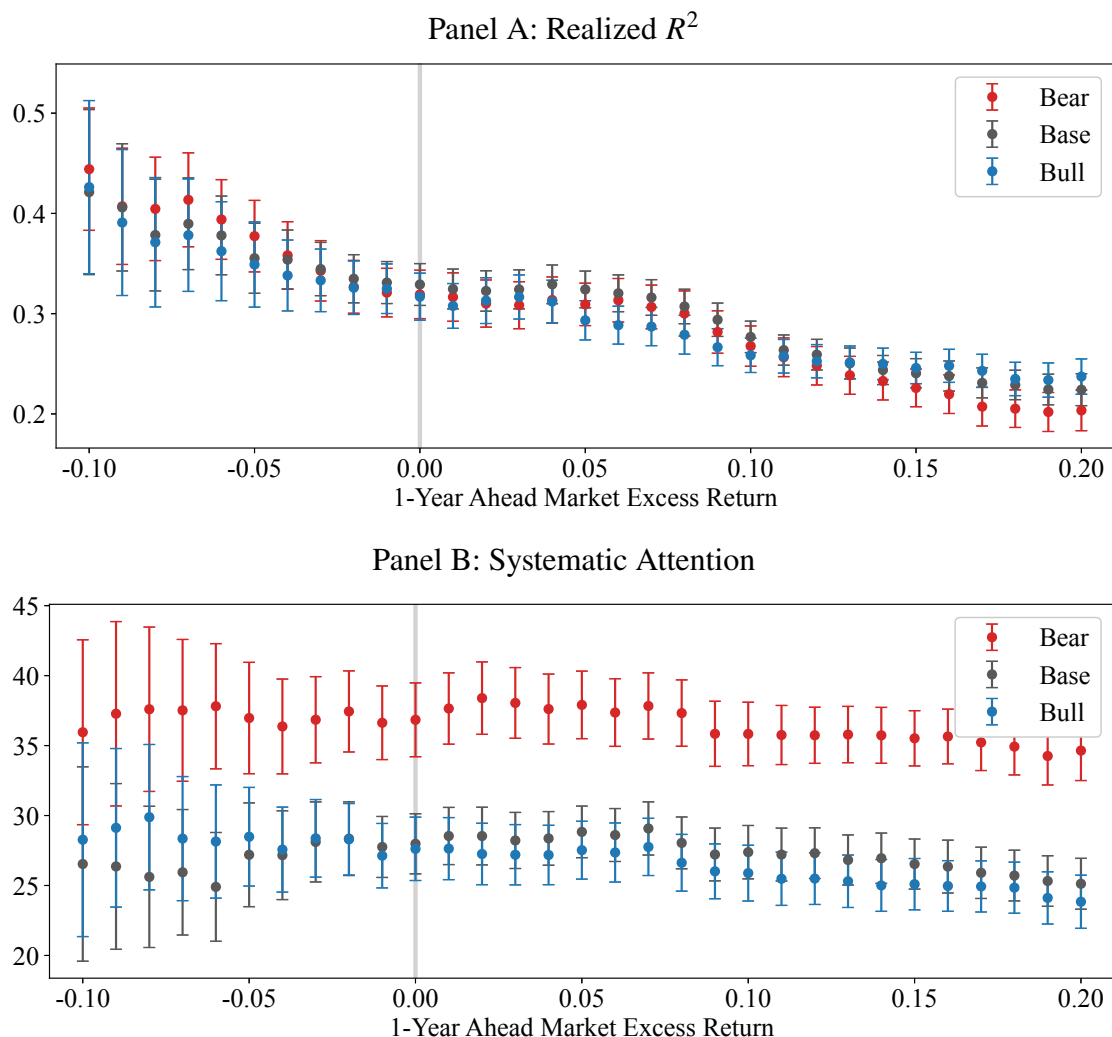


Figure A.3 Realized R^2 and Systematic Attention Conditional on Market States.

Note: This figure shows the realized R^2 and systematic attention conditional on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Systematic attention is measured as the firm-level average systematic attention within each realized scenario (bear, base, and bull). Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear/base/bull cases. Error bars show 95% confidence intervals from constant-only regressions.

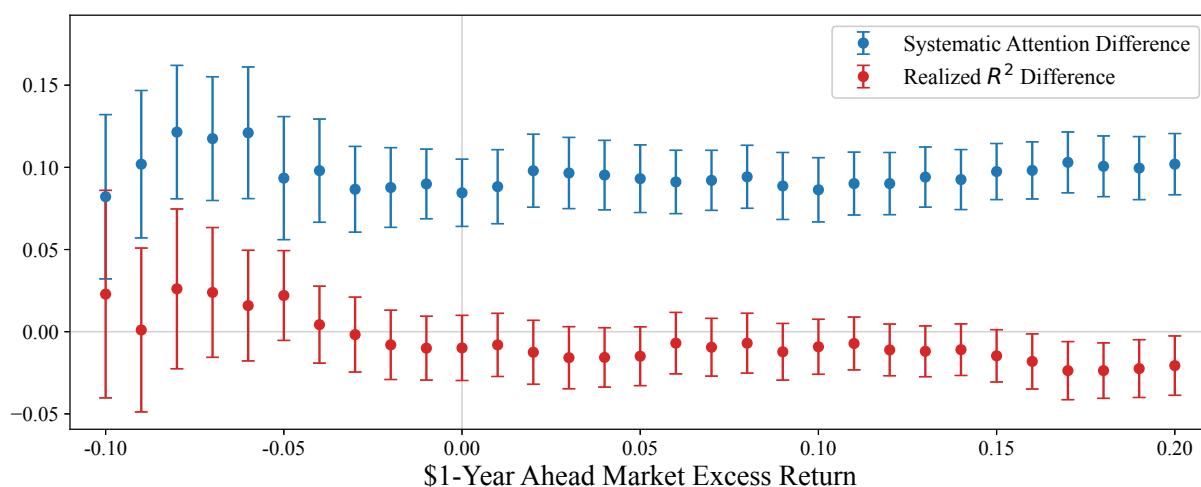


Figure A.4 Bear – Base Gaps in Systematic Attention and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bear–base differences in systematic attention and realized CAPM R^2 on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Systematic attention is measured as the firm-level average systematic attention in the bear and base cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear or realized base cases. The plotted values represent the bear-minus-base differences. Error bars show 95% confidence intervals from constant-only regressions.

Table A.1 Descriptions of Key Topics Used in Attention and Correlation Analysis

Topic	Description	Bear Attn. (%)	News Attn. (%)	News Sent.
Economic Growth and Recovery Outlook	Indicators suggesting potential economic recovery emerge from consumer confidence, job growth, and corporate earnings. Positive signals improve investor outlook, although inflation and labor market weakness may remain concerns.	10.66	2.36	0.48
COVID-19 Pandemic and Vaccine Developments	The COVID-19 pandemic significantly affected global health and economic conditions. Vaccine development, distribution logistics, variants, and public health responses shaped market expectations and business conditions.	5.36	5.34	0.41
Market Activity and Financial Performance	Stock market activity reacts to earnings releases, firm performance, and sector-wide developments. Strong results support rallies, while weak performance increases caution and volatility.	4.75	5.44	0.39
Consumer Price Index and Inflation Trends	Rising consumer prices and inflation pressures influence monetary policy, investor sentiment, and broader economic strategies. Central bank responses are especially important for markets.	2.40	0.86	0.08
International Trade Relations and Economic Policies	Trade tensions, tariff policy, and negotiations among major economies affect global commerce, supply chains, and investor confidence.	2.31	1.60	0.12
Consumer Behavior Amid Economic Strains	Economic pressures change household spending patterns, especially for discretionary purchases. Firms adjust pricing and marketing strategies amid inflation and fuel cost concerns.	1.91	0.24	-0.05
Geopolitical Tensions and Economic Impacts	Political instability, wars, and international conflicts affect stock prices, commodity markets, and investor sentiment through heightened uncertainty.	0.46	4.40	-0.11
Economic Stimulus and Government Interventions	Fiscal stimulus packages, government support programs, and intervention policies are used to stabilize growth, employment, and financial markets during downturns.	0.40	0.46	0.07

Table A.3 Regression Results (Bull Scenario, Spec 3, Time FE)

	$\tau = 1$	$\tau = 2$	$\tau = 3$	$\tau = 4$
x_1	0.0315* (1.85)	0.0250* (1.70)	0.0103 (0.97)	-0.0025 (-0.35)
x_2	-0.0783 (-1.35)	-0.0876 (-1.52)	-0.0540 (-1.08)	-0.0338 (-0.83)
x_3	0.0003** (2.08)	0.0003* (1.85)	0.0002 (1.29)	0.0003** (1.97)
Attention Lag 1	0.2889*** (11.22)	0.2849*** (10.95)	0.2842*** (10.86)	0.2849*** (10.85)
Attention Lag 2	0.1844*** (6.65)	0.1864*** (6.70)	0.1818*** (6.51)	0.1804*** (6.45)
Attention Lag 3	0.4919*** (16.22)	0.4938*** (16.30)	0.4987*** (16.39)	0.5004*** (16.37)
Observations	11440	11388	11336	11284
R^2	0.8515	0.8520	0.8526	0.8536
Time FE	Yes	Yes	Yes	Yes

t-statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A.4 Analyst Experience and Macro Bear Bias

Note: This table reports panel regressions of MacroGap (bear minus base macro attention, 0–100 scale) on analyst experience measures. Junior (MS) equals one if the analyst’s tenure at Morgan Stanley is at most 36 months. Junior (Profession) equals one if the analyst’s industry experience across all brokerages is at most 3 years (identified from the full IBES price-target detail file). Columns (3)–(4) use continuous experience measures in years. Column (5) includes both MS and profession experience simultaneously. All columns are restricted to the analyst-matched subsample. All specifications include firm and year fixed effects. Standard errors are clustered by firm. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	MacroGap				
	(1)	(2)	(3)	(4)	(5)
	Junior (MS)	Junior (Prof.)	Exp. MS	Exp. Prof.	Horse Race
Junior (MS)	−0.6924* (−1.66)				
Junior (Profession)		0.0759 (0.15)			
Experience MS (years)			0.0750 (0.94)		0.0914 (0.79)
Experience Prof. (years)				0.0223 (0.48)	−0.0142 (−0.21)
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Observations	17,257	17,257	17,257	17,257	17,257
R^2	0.0005	0.0000	0.0003	0.0001	0.0003

B.1 Additional Evidence on Mechanisms

Table A.4 reports panel regressions of MacroGap on analyst experience, measured as Morgan Stanley tenure (columns 1 and 3), industry-wide experience across all brokerages (columns 2 and 4), and a horse-race specification including both measures (column 5). See Section 4.2 for discussion.

Table A.5 compares the within- R^2 of a baseline specification (firm and year fixed effects) with a specification that adds 232 analyst fixed effects. See Section 4.2 for discussion.

Table A.5 Analyst Fixed Effects and Macro Bear Bias

Note: This table reports the incremental R^2 from adding analyst fixed effects to a specification with firm and year fixed effects. The dependent variable is MacroGap (bear minus base macro attention, 0–100 scale). Firm effects are absorbed by within-firm demeaning. Year effects are captured by year dummies. Model (B) adds 232 analyst dummies. Both models are estimated on the analyst-matched subsample. ΔR^2 is the incremental R^2 from adding analyst fixed effects.

	(A)	(B)
	Baseline	+ Analyst FE
R^2	0.0028	0.0233
ΔR^2		0.0205
Firm FE	✓	✓
Year FE	✓	✓
Analyst FE		✓
Observations	17,257	17,257
Firms	1,274	1,274
Analysts		232